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Inventor(s)	Adie; Gordon Campbell et al.

Wound dressing and method of use

Abstract

A system, method, and apparatus are disclosed for dressing a wound. The apparatus comprises a liquid and gas permeable transmission layer, an absorbent layer for absorbing wound exudate, the absorbent layer overlying the transmission layer, a gas impermeable cover layer overlying the absorbent layer and comprising a first orifice, wherein the cover layer is moisture vapor permeable.

Inventors: **Adie; Gordon Campbell (Cottingham, GB), Collinson; Sarah Jenny (Hull, GB), Fryer; Christopher John (York, GB), Hartwell; Edward Yerbury (Hull, GB), Nicolini; Derek (Hull, GB), Peron; Yannick Louis (East Yorkshire, GB)**

Applicant: **Smith & Nephew PLC** (Watford, GB)

Family ID: **1000008818383**

Assignee: **Smith & Nephew PLC (Watford, GB)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2280915	12/1941	Johnson	N/A	N/A
2568933	12/1950	Robbins	N/A	N/A
2632443	12/1952	Lesher	N/A	N/A
2682873	12/1953	Idnis et al.	N/A	N/A
2877765	12/1958	John et al.	N/A	N/A
2889039	12/1958	Peter et al.	N/A	N/A
2923298	12/1959	Dockstader et al.	N/A	N/A
3073304	12/1962	Schaar	N/A	N/A
3115138	12/1962	McElvenny et al.	N/A	N/A
3367332	12/1967	Groves	N/A	N/A
3572340	12/1970	Lloyd et al.	N/A	N/A
3874387	12/1974	Barbieri	N/A	N/A
3903882	12/1974	Augurt	N/A	N/A
3972328	12/1975	Chen	N/A	N/A
3993080	12/1975	Loseff	N/A	N/A
4029598	12/1976	Neisius et al.	N/A	N/A
4112947	12/1977	Nehring	N/A	N/A
4117551	12/1977	Brooks et al.	N/A	N/A
4136696	12/1978	Nehring	N/A	N/A
4181127	12/1979	Linsky	128/889	A61L 15/24
4184510	12/1979	Murry et al.	N/A	N/A
4217894	12/1979	Franetzki	N/A	N/A
4219019	12/1979	Coates	N/A	N/A
4224945	12/1979	Cohen	N/A	N/A
4294240	12/1980	Thill	N/A	N/A

4316466	12/1981	Babb	N/A	N/A
4382441	12/1982	Svedman	N/A	N/A
4465485	12/1983	Kashmer et al.	N/A	N/A
4534356	12/1984	Papadakis	N/A	N/A
4538920	12/1984	Drake	N/A	N/A
4541426	12/1984	Webster	N/A	N/A
4573965	12/1985	Russo	N/A	N/A
4592751	12/1985	Gegelys	N/A	N/A
4655754	12/1986	Richmond et al.	N/A	N/A
4655766	12/1986	Theeuwes et al.	N/A	N/A
4710165	12/1986	McNeil et al.	N/A	N/A
4728499	12/1987	Fehder	N/A	N/A
4753536	12/1987	Spehar et al.	N/A	N/A
4767026	12/1987	Keller et al.	N/A	N/A
4771919	12/1987	Ernst	N/A	N/A
4778446	12/1987	Jensen	N/A	N/A
4792328	12/1987	Beck et al.	N/A	N/A
4813942	12/1988	Alvarez	N/A	N/A
4826494	12/1988	Richmond et al.	N/A	N/A
4828546	12/1988	McNeil et al.	N/A	N/A
4846164	12/1988	Martz	N/A	N/A
4872450	12/1988	Austad	N/A	N/A
4921488	12/1989	Maitz et al.	N/A	N/A
4929477	12/1989	Will	N/A	N/A
4936834	12/1989	Beck et al.	N/A	N/A
4969880	12/1989	Zamierowski	N/A	N/A
4972829	12/1989	Knerr	N/A	N/A
4994022	12/1990	Steffler et al.	N/A	N/A
5010115	12/1990	Grisoni	N/A	N/A
5010883	12/1990	Rawlings et al.	N/A	N/A
5055198	12/1990	Shettigar	N/A	N/A
5056510	12/1990	Gilman	N/A	N/A
5064653	12/1990	Sessions et al.	N/A	N/A
5073172	12/1990	Fell	N/A	N/A
5080493	12/1991	McKown et al.	N/A	N/A
5100396	12/1991	Zamierowski	N/A	N/A
5115801	12/1991	Cartmell et al.	N/A	N/A
5134994	12/1991	Say	N/A	N/A
5145906	12/1991	Chambers et al.	N/A	N/A
5152757	12/1991	Eriksson	N/A	N/A
5160328	12/1991	Cartmell et al.	N/A	N/A
5167613	12/1991	Karami et al.	N/A	N/A
5176663	12/1992	Svedman et al.	N/A	N/A
5181905	12/1992	Flam	N/A	N/A
5197945	12/1992	Cole et al.	N/A	N/A
5215519	12/1992	Shettigar	N/A	N/A
5234419	12/1992	Bryant et al.	N/A	N/A
5238732	12/1992	Krishnan	N/A	N/A
5249709	12/1992	Duckworth et al.	N/A	N/A
5261893	12/1992	Zamierowski	N/A	N/A
5266326	12/1992	Barry et al.	N/A	N/A
5266928	12/1992	Johnson	N/A	N/A
5267952	12/1992	Gardner	N/A	N/A
5279608	12/1993	Cherif Cheikh	N/A	N/A
5308313	12/1993	Karami et al.	N/A	N/A

5328614	12/1993	Matsumura	N/A	N/A
5333760	12/1993	Simmen	N/A	N/A
5336219	12/1993	Krantz	N/A	N/A
5354261	12/1993	Clark et al.	N/A	N/A
5358494	12/1993	Svedman	N/A	N/A
5364381	12/1993	Soga et al.	N/A	N/A
5380280	12/1994	Peterson	N/A	N/A
5380294	12/1994	Persson	N/A	N/A
5397316	12/1994	LaVon et al.	N/A	N/A
5437651	12/1994	Todd et al.	N/A	N/A
5445604	12/1994	Lang	N/A	N/A
5447492	12/1994	Cartmell et al.	N/A	N/A
5456660	12/1994	Reich et al.	N/A	N/A
5456745	12/1994	Roreger et al.	N/A	N/A
5480377	12/1995	Cartmell et al.	N/A	N/A
5489280	12/1995	Russell	N/A	N/A
5497788	12/1995	Inman et al.	N/A	N/A
5498338	12/1995	Kruger et al.	N/A	N/A
5527293	12/1995	Zamierowski	N/A	N/A
5538500	12/1995	Peterson	N/A	N/A
5540922	12/1995	Fabo	N/A	N/A
5549584	12/1995	Gross	N/A	N/A
5553998	12/1995	Muhlhoff et al.	N/A	N/A
5562107	12/1995	Lavender et al.	N/A	N/A
5579765	12/1995	Cox et al.	N/A	N/A
5582596	12/1995	Fukunaga et al.	N/A	N/A
5583114	12/1995	Barrows et al.	N/A	N/A
5603946	12/1996	Constantine	N/A	N/A
5609271	12/1996	Keller et al.	N/A	N/A
5612050	12/1996	Rowe et al.	N/A	N/A
5624423	12/1996	Anjur et al.	N/A	N/A
5636643	12/1996	Argenta et al.	N/A	N/A
5637080	12/1996	Geng	N/A	N/A
5643189	12/1996	Masini	N/A	N/A
5645081	12/1996	Argenta et al.	N/A	N/A
5660823	12/1996	Chakrabarti et al.	N/A	N/A
5662599	12/1996	Reich et al.	N/A	N/A
5702356	12/1996	Hathman	N/A	N/A
5704905	12/1997	Jensen et al.	N/A	N/A
5707499	12/1997	Joshi et al.	N/A	N/A
5713384	12/1997	Roach et al.	N/A	N/A
5717030	12/1997	Dunn et al.	N/A	N/A
5747064	12/1997	Burnett et al.	N/A	N/A
5759570	12/1997	Arnold	N/A	N/A
5776193	12/1997	Kwan et al.	N/A	N/A
5785688	12/1997	Joshi et al.	N/A	N/A
5795439	12/1997	Euripides et al.	N/A	N/A
5804021	12/1997	Abuto et al.	N/A	N/A
5827213	12/1997	Jensen	N/A	N/A
5833646	12/1997	Masini	N/A	N/A
5834007	12/1997	Kubota	N/A	N/A
5840052	12/1997	Johns	N/A	N/A
5843011	12/1997	Lucas	N/A	N/A
5843025	12/1997	Shaari	N/A	N/A
5852126	12/1997	Barnard et al.	N/A	N/A

5868724	12/1998	Dierckes, Jr. et al.	N/A	N/A
5868933	12/1998	Patrick et al.	N/A	N/A
5874500	12/1998	Rhee et al.	N/A	N/A
5876611	12/1998	Shettigar	N/A	N/A
5897541	12/1998	Uitenbroek et al.	N/A	N/A
5945115	12/1998	Dunn et al.	N/A	N/A
5962010	12/1998	Greff et al.	N/A	N/A
5964723	12/1998	Augustine	N/A	N/A
5981822	12/1998	Addison	N/A	N/A
5998472	12/1998	Berger et al.	N/A	N/A
6040493	12/1999	Cooke et al.	N/A	N/A
6071267	12/1999	Zamierowski	N/A	N/A
6075177	12/1999	Bahia et al.	N/A	N/A
6103951	12/1999	Freeman	N/A	N/A
6124520	12/1999	Roberts	N/A	N/A
6124521	12/1999	Roberts	N/A	N/A
6135116	12/1999	Vogel et al.	N/A	N/A
6142982	12/1999	Hunt et al.	N/A	N/A
6143352	12/1999	Clark et al.	N/A	N/A
6165201	12/1999	Sawhney et al.	N/A	N/A
6167613	12/2000	Jarrett et al.	N/A	N/A
6168788	12/2000	Wortham	N/A	N/A
6168800	12/2000	Dobos et al.	N/A	N/A
6203654	12/2000	McFall et al.	N/A	N/A
6207875	12/2000	Lindqvist et al.	N/A	N/A
6214332	12/2000	Askill et al.	N/A	N/A
6225523	12/2000	Masini	N/A	N/A
6252129	12/2000	Coffee	N/A	N/A
6254567	12/2000	Treu et al.	N/A	N/A
6255552	12/2000	Cummings et al.	N/A	N/A
6287521	12/2000	Quay et al.	N/A	N/A
6291050	12/2000	Cree et al.	N/A	N/A
6297422	12/2000	Hansen et al.	N/A	N/A
6297423	12/2000	Schoenfeldt et al.	N/A	N/A
6345623	12/2001	Heaton et al.	N/A	N/A
6362390	12/2001	Carlucci et al.	N/A	N/A
6391294	12/2001	Dettmar et al.	N/A	N/A
6398761	12/2001	Bills et al.	N/A	N/A
6398767	12/2001	Fleischmann	N/A	N/A
6440167	12/2001	Shimizu	N/A	N/A
6447802	12/2001	Sessions et al.	N/A	N/A
6450773	12/2001	Upton	N/A	N/A
6458109	12/2001	Henley et al.	N/A	N/A
6465708	12/2001	Augustine	N/A	N/A
6468295	12/2001	Augustine et al.	N/A	N/A
6471982	12/2001	Lydon et al.	N/A	N/A
6486285	12/2001	Fujita	N/A	N/A
6491684	12/2001	Joshi et al.	N/A	N/A
6495127	12/2001	Wallace et al.	N/A	N/A
6506175	12/2002	Goldstein	N/A	N/A
6509031	12/2002	Miller et al.	N/A	N/A
6521251	12/2002	Askill et al.	N/A	N/A
6528696	12/2002	Ireland	N/A	N/A
6547467	12/2002	Quintero	N/A	N/A
6553998	12/2002	Heaton et al.	N/A	N/A

6566575	12/2002	Stickels et al.	N/A	N/A
6575940	12/2002	Levinson et al.	N/A	N/A
6586653	12/2002	Graeme, III et al.	N/A	N/A
6596704	12/2002	Court et al.	N/A	N/A
6599262	12/2002	Masini	N/A	N/A
6613953	12/2002	Altura	N/A	N/A
6626891	12/2002	Ohmstede	N/A	N/A
6627216	12/2002	Brandt et al.	N/A	N/A
6629774	12/2002	Gruendeman	N/A	N/A
6648862	12/2002	Watson	N/A	N/A
6653520	12/2002	Mouton	N/A	N/A
6673028	12/2003	Argenta et al.	N/A	N/A
6685681	12/2003	Lockwood et al.	N/A	N/A
6695823	12/2003	Lina et al.	N/A	N/A
6695824	12/2003	Howard et al.	N/A	N/A
6706940	12/2003	Worthley	N/A	N/A
6719742	12/2003	McCormack et al.	N/A	N/A
6730299	12/2003	Tayot et al.	N/A	N/A
6752794	12/2003	Lockwood et al.	N/A	N/A
6755807	12/2003	Risk, Jr. et al.	N/A	N/A
6762337	12/2003	Boukanov et al.	N/A	N/A
6764459	12/2003	Donaldson	N/A	N/A
6764462	12/2003	Risk, Jr. et al.	N/A	N/A
6767334	12/2003	Randolph	N/A	N/A
6775807	12/2003	Lowther et al.	N/A	N/A
6776769	12/2003	Smith	N/A	N/A
6787682	12/2003	Gilman	N/A	N/A
6790438	12/2003	Constancis et al.	N/A	N/A
6814079	12/2003	Heaton et al.	N/A	N/A
6824533	12/2003	Risk, Jr. et al.	N/A	N/A
6838589	12/2004	Liedtke et al.	N/A	N/A
6855135	12/2004	Lockwood et al.	N/A	N/A
6856821	12/2004	Johnson	N/A	N/A
6936037	12/2004	Bubb et al.	N/A	N/A
6951553	12/2004	Bubb et al.	N/A	N/A
6960179	12/2004	Gura	N/A	N/A
6977323	12/2004	Swenson	N/A	N/A
6979324	12/2004	Bybordi et al.	N/A	N/A
6994702	12/2005	Johnson	N/A	N/A
6998511	12/2005	Worthley	N/A	N/A
7004915	12/2005	Boynton et al.	N/A	N/A
7022113	12/2005	Lockwood et al.	N/A	N/A
7049478	12/2005	Smith	N/A	N/A
7067709	12/2005	Murata et al.	N/A	N/A
7070584	12/2005	Johnson et al.	N/A	N/A
7077832	12/2005	Fleischmann	N/A	N/A
7087806	12/2005	Scheinberg et al.	N/A	N/A
7108683	12/2005	Zamierowski	N/A	N/A
7117869	12/2005	Heaton et al.	N/A	N/A
7118545	12/2005	Boyde	N/A	N/A
7128735	12/2005	Weston	N/A	N/A
7129210	12/2005	Lowinger et al.	N/A	N/A
7195624	12/2006	Lockwood et al.	N/A	N/A
7214202	12/2006	Vogel et al.	N/A	N/A
7216651	12/2006	Argenta et al.	N/A	N/A

7238850	12/2006	Shimanuki	N/A	N/A
7294751	12/2006	Propp et al.	N/A	N/A
7294752	12/2006	Propp	N/A	N/A
7303757	12/2006	Schankereli et al.	N/A	N/A
7335809	12/2007	Riesinger	N/A	N/A
7338482	12/2007	Lockwood et al.	N/A	N/A
7361184	12/2007	Joshi	N/A	N/A
7381211	12/2007	Zamierowski	N/A	N/A
7381859	12/2007	Hunt et al.	N/A	N/A
7401394	12/2007	Muller	N/A	N/A
7429689	12/2007	Chen et al.	N/A	N/A
7438705	12/2007	Karpowicz et al.	N/A	N/A
7468471	12/2007	Sigurjonsson et al.	N/A	N/A
7470830	12/2007	Sigurjonsson et al.	N/A	N/A
7476205	12/2008	Erdmann	N/A	N/A
7485112	12/2008	Karpowicz et al.	N/A	N/A
7494482	12/2008	Orgill et al.	N/A	N/A
7503910	12/2008	Adahan	N/A	N/A
7507870	12/2008	Nielsen et al.	N/A	N/A
7511187	12/2008	Kelly	N/A	N/A
7524286	12/2008	Johnson	N/A	N/A
7524315	12/2008	Blott et al.	N/A	N/A
7534240	12/2008	Johnson	N/A	N/A
7534927	12/2008	Lockwood et al.	N/A	N/A
7563940	12/2008	Kurata	N/A	N/A
7569742	12/2008	Haggstrom et al.	N/A	N/A
7576256	12/2008	Bjornberg et al.	N/A	N/A
7601129	12/2008	Aali	N/A	N/A
7605298	12/2008	Bechert et al.	N/A	N/A
7612247	12/2008	Oyaski	N/A	N/A
7615036	12/2008	Joshi et al.	N/A	N/A
7622629	12/2008	Aali	N/A	N/A
7625362	12/2008	Boehringer et al.	N/A	N/A
7645269	12/2009	Zamierowski	N/A	N/A
7676400	12/2009	Dillon	N/A	N/A
7699823	12/2009	Haggstrom et al.	N/A	N/A
7699830	12/2009	Martin	N/A	N/A
7700819	12/2009	Ambrosio et al.	N/A	N/A
7708724	12/2009	Weston	N/A	N/A
7717313	12/2009	Criscuolo et al.	N/A	N/A
7718249	12/2009	Russell et al.	N/A	N/A
7722582	12/2009	Lina et al.	N/A	N/A
7723560	12/2009	Lockwood et al.	N/A	N/A
7723561	12/2009	Propp	N/A	N/A
7731702	12/2009	Bybordi et al.	N/A	N/A
7745681	12/2009	Ferguson	N/A	N/A
7749531	12/2009	Booher	N/A	N/A
7753894	12/2009	Blott et al.	N/A	N/A
7759537	12/2009	Bishop et al.	N/A	N/A
7759538	12/2009	Fleischmann	N/A	N/A
7759539	12/2009	Shaw et al.	N/A	N/A
7775998	12/2009	Riesinger	N/A	N/A
7776028	12/2009	Miller et al.	N/A	N/A
7779625	12/2009	Joshi et al.	N/A	N/A
7790945	12/2009	Watson, Jr.	N/A	N/A

7794438	12/2009	Henley et al.	N/A	N/A
7794450	12/2009	Blott et al.	N/A	N/A
7803980	12/2009	Griffiths et al.	N/A	N/A
7811269	12/2009	Boynton et al.	N/A	N/A
7815616	12/2009	Boehringer et al.	N/A	N/A
7816577	12/2009	Aali	N/A	N/A
7828782	12/2009	Suzuki	N/A	N/A
7838716	12/2009	De Luis et al.	N/A	N/A
7838717	12/2009	Haggstrom et al.	N/A	N/A
7846141	12/2009	Weston	N/A	N/A
7857806	12/2009	Karpowicz et al.	N/A	N/A
7858838	12/2009	Holm et al.	N/A	N/A
7862831	12/2010	Wang et al.	N/A	N/A
7880050	12/2010	Robinson et al.	N/A	N/A
7884258	12/2010	Boehringer et al.	N/A	N/A
7896823	12/2010	Mangrum et al.	N/A	N/A
7896856	12/2010	Petrosenko et al.	N/A	N/A
7909805	12/2010	Weston	N/A	N/A
7910135	12/2010	St. John et al.	N/A	N/A
7910791	12/2010	Coffey	N/A	N/A
7922703	12/2010	Riesinger	N/A	N/A
7927318	12/2010	Risk, Jr. et al.	N/A	N/A
7931630	12/2010	Nishtala et al.	N/A	N/A
7935066	12/2010	Shives et al.	N/A	N/A
7959624	12/2010	Riesinger	N/A	N/A
7964766	12/2010	Blott et al.	N/A	N/A
7976519	12/2010	Bubb et al.	N/A	N/A
7982087	12/2010	Greener et al.	N/A	N/A
7985209	12/2010	Villanueva et al.	N/A	N/A
D642594	12/2010	Mattson et al.	N/A	N/A
7988673	12/2010	Wright et al.	N/A	N/A
7998125	12/2010	Weston	N/A	N/A
8007164	12/2010	Miyano et al.	N/A	N/A
8007257	12/2010	Heaton et al.	N/A	N/A
8021347	12/2010	Vitaris et al.	N/A	N/A
8025650	12/2010	Anderson et al.	N/A	N/A
8034037	12/2010	Adams et al.	N/A	N/A
8062272	12/2010	Weston	N/A	N/A
8062331	12/2010	Zamierowski	N/A	N/A
8080702	12/2010	Blott et al.	N/A	N/A
8084663	12/2010	Watson, Jr.	N/A	N/A
8092436	12/2011	Christensen	N/A	N/A
8097272	12/2011	Addison	N/A	N/A
8100887	12/2011	Weston et al.	N/A	N/A
8105295	12/2011	Blott et al.	N/A	N/A
8118794	12/2011	Weston	N/A	N/A
8119160	12/2011	Looney et al.	N/A	N/A
8133211	12/2011	Cavanaugh, II et al.	N/A	N/A
8147468	12/2011	Barta et al.	N/A	N/A
8152785	12/2011	Vitaris	N/A	N/A
8158844	12/2011	McNeil	N/A	N/A
8162907	12/2011	Heagle	N/A	N/A
8162909	12/2011	Blott et al.	N/A	N/A
8168848	12/2011	Lockwood et al.	N/A	N/A
8188331	12/2011	Barta et al.	N/A	N/A

8202261	12/2011	Kazala, Jr. et al.	N/A	N/A
8207392	12/2011	Haggstrom et al.	N/A	N/A
8211071	12/2011	Mormino et al.	N/A	N/A
8212101	12/2011	Propp	N/A	N/A
8215929	12/2011	Shen et al.	N/A	N/A
8226942	12/2011	Charier et al.	N/A	N/A
8235955	12/2011	Blott et al.	N/A	N/A
8235972	12/2011	Adahan	N/A	N/A
8241261	12/2011	Randolph et al.	N/A	N/A
8246607	12/2011	Karpowicz et al.	N/A	N/A
8251979	12/2011	Malhi	N/A	N/A
8252971	12/2011	Aali et al.	N/A	N/A
8257327	12/2011	Blott et al.	N/A	N/A
8257328	12/2011	Augustine et al.	N/A	N/A
8267908	12/2011	Coulthard	N/A	N/A
8273368	12/2011	Ambrosio et al.	N/A	N/A
8282611	12/2011	Weston	N/A	N/A
8294586	12/2011	Pidgeon et al.	N/A	N/A
8303552	12/2011	Weston	N/A	N/A
8308714	12/2011	Weston et al.	N/A	N/A
8314283	12/2011	Kingsford et al.	N/A	N/A
8317774	12/2011	Adahan	N/A	N/A
8323264	12/2011	Weston et al.	N/A	N/A
8328858	12/2011	Barsky et al.	N/A	N/A
8348910	12/2012	Blott et al.	N/A	N/A
8372049	12/2012	Jaeb et al.	N/A	N/A
8372050	12/2012	Jaeb et al.	N/A	N/A
8376972	12/2012	Fleischmann	N/A	N/A
8382731	12/2012	Johannison	N/A	N/A
8403899	12/2012	Sherman	N/A	N/A
8404921	12/2012	Lee et al.	N/A	N/A
D679819	12/2012	Peron	N/A	N/A
D679820	12/2012	Peron	N/A	N/A
8409159	12/2012	Hu et al.	N/A	N/A
8410189	12/2012	Carnahan et al.	N/A	N/A
8425478	12/2012	Olson	N/A	N/A
8444612	12/2012	Patel et al.	N/A	N/A
8449509	12/2012	Weston	N/A	N/A
8460255	12/2012	Joshi et al.	N/A	N/A
8494349	12/2012	Gordon	N/A	N/A
8506554	12/2012	Adahan	N/A	N/A
8535283	12/2012	Heaton et al.	N/A	N/A
8535296	12/2012	Blott et al.	N/A	N/A
8540687	12/2012	Henley et al.	N/A	N/A
8545466	12/2012	Andresen et al.	N/A	N/A
8556871	12/2012	Mormino et al.	N/A	N/A
8568386	12/2012	Malhi	N/A	N/A
8569566	12/2012	Blott et al.	N/A	N/A
8624077	12/2013	Rosenberg	N/A	N/A
8628505	12/2013	Weston	N/A	N/A
8641691	12/2013	Fink et al.	N/A	N/A
8663198	12/2013	Buan et al.	N/A	N/A
8679079	12/2013	Heaton et al.	N/A	N/A
8680360	12/2013	Greener et al.	N/A	N/A
8715256	12/2013	Greener	N/A	N/A

8734410	12/2013	Hall et al.	N/A	N/A
8753670	12/2013	Delmotte	N/A	N/A
8764732	12/2013	Hartwell	N/A	N/A
8784392	12/2013	Vess et al.	N/A	N/A
8791316	12/2013	Greener	N/A	N/A
8795243	12/2013	Weston	N/A	N/A
8795244	12/2013	Randolph et al.	N/A	N/A
8795247	12/2013	Bennett et al.	N/A	N/A
8795635	12/2013	Tamarkin et al.	N/A	N/A
8795713	12/2013	Makower et al.	N/A	N/A
8795800	12/2013	Evans	N/A	N/A
8801685	12/2013	Armstrong et al.	N/A	N/A
8808274	12/2013	Hartwell	N/A	N/A
8814842	12/2013	Coulthard et al.	N/A	N/A
8829263	12/2013	Haggstrom et al.	N/A	N/A
8834451	12/2013	Blott et al.	N/A	N/A
8834452	12/2013	Hudspeth et al.	N/A	N/A
8843327	12/2013	Vernon-Harcourt et al.	N/A	N/A
8864748	12/2013	Coulthard et al.	N/A	N/A
8915895	12/2013	Jaeb et al.	N/A	N/A
8916742	12/2013	Smith	N/A	N/A
8956336	12/2014	Haggstrom et al.	N/A	N/A
8968773	12/2014	Thomas et al.	N/A	N/A
9012714	12/2014	Fleischmann	N/A	N/A
9028872	12/2014	Gaserod et al.	N/A	N/A
9033942	12/2014	Vess	N/A	N/A
9044579	12/2014	Blott et al.	N/A	N/A
9061095	12/2014	Adie et al.	N/A	N/A
9168330	12/2014	Joshi et al.	N/A	N/A
9180231	12/2014	Greener	N/A	N/A
9198801	12/2014	Weston	N/A	N/A
9199012	12/2014	Vitaris et al.	N/A	N/A
9220822	12/2014	Hartwell	N/A	N/A
9302033	12/2015	Riesinger	N/A	N/A
9375353	12/2015	Vitaris et al.	N/A	N/A
9375521	12/2015	Hudspeth et al.	N/A	N/A
9381283	12/2015	Adams et al.	N/A	N/A
9421309	12/2015	Robinson et al.	N/A	N/A
9446178	12/2015	Blott et al.	N/A	N/A
9452248	12/2015	Blott et al.	N/A	N/A
9629986	12/2016	Patel et al.	N/A	N/A
9820888	12/2016	Greener et al.	N/A	N/A
10080689	12/2017	Hall et al.	N/A	N/A
2001/0004082	12/2000	Keller et al.	N/A	N/A
2001/0034499	12/2000	Sessions et al.	N/A	N/A
2001/0043913	12/2000	Spaans et al.	N/A	N/A
2001/0043943	12/2000	Coffey	N/A	N/A
2001/0051178	12/2000	Blatchford et al.	N/A	N/A
2002/0002209	12/2001	Mork	N/A	N/A
2002/0019602	12/2001	Geng	N/A	N/A
2002/0038826	12/2001	Hurray et al.	N/A	N/A
2002/0052570	12/2001	Naimer	N/A	N/A
2002/0087136	12/2001	Widlund	N/A	N/A
2002/0110672	12/2001	Muratore-Pallatino et al.	N/A	N/A
2002/0122771	12/2001	Holland et al.	N/A	N/A

2002/0143286	12/2001	Tumey	N/A	N/A
2002/0145007	12/2001	Sawhney et al.	N/A	N/A
2002/0169405	12/2001	Roberts	N/A	N/A
2002/0183702	12/2001	Henley et al.	N/A	N/A
2002/0187182	12/2001	Kramer et al.	N/A	N/A
2002/0198490	12/2001	Wirt et al.	N/A	N/A
2003/0009122	12/2002	Veras	N/A	N/A
2003/0021775	12/2002	Freeman	N/A	N/A
2003/0040478	12/2002	Drucker et al.	N/A	N/A
2003/0045825	12/2002	Etheredge, III	602/45	A61F 13/0226
2003/0050590	12/2002	Kirsch	N/A	N/A
2003/0069535	12/2002	Shalaby	N/A	N/A
2003/0069563	12/2002	Johnson	N/A	N/A
2003/0114818	12/2002	Benecke et al.	N/A	N/A
2003/0125646	12/2002	Whitlock	N/A	N/A
2003/0125649	12/2002	McIntosh et al.	N/A	N/A
2003/0143189	12/2002	Askill et al.	N/A	N/A
2003/0144619	12/2002	Augustine	N/A	N/A
2003/0153860	12/2002	Nielsen	602/58	A61F 13/0259
2003/0171675	12/2002	Rosenberg	N/A	N/A
2003/0175798	12/2002	Raees et al.	N/A	N/A
2003/0180341	12/2002	Gooch et al.	N/A	N/A
2003/0183653	12/2002	Bills	N/A	N/A
2003/0199800	12/2002	Levin	N/A	N/A
2003/0203011	12/2002	Abuelyaman et al.	N/A	N/A
2003/0212357	12/2002	Pace	N/A	N/A
2003/0212359	12/2002	Butler	N/A	N/A
2003/0212431	12/2002	Brady et al.	N/A	N/A
2003/0219469	12/2002	Johnson et al.	N/A	N/A
2003/0225347	12/2002	Argenta et al.	N/A	N/A
2004/0019337	12/2003	Moberg-Alehammar et al.	N/A	N/A
2004/0019342	12/2003	Nagasuna et al.	N/A	N/A
2004/0033466	12/2003	Shellard et al.	N/A	N/A
2004/0037897	12/2003	Benjamin et al.	N/A	N/A
2004/0039391	12/2003	Argenta et al.	N/A	N/A
2004/0049187	12/2003	Burnett et al.	N/A	N/A
2004/0054338	12/2003	Bybordi et al.	N/A	N/A
2004/0057855	12/2003	Gerlach et al.	N/A	N/A
2004/0064111	12/2003	Lockwood et al.	N/A	N/A
2004/0073152	12/2003	Karason et al.	N/A	N/A
2004/0078011	12/2003	Stevens	N/A	N/A
2004/0106888	12/2003	Lutri et al.	N/A	N/A
2004/0122434	12/2003	Argenta et al.	N/A	N/A
2004/0127863	12/2003	Bubb et al.	N/A	N/A
2004/0138602	12/2003	Rossen	N/A	N/A
2004/0167617	12/2003	Voellmicke et al.	N/A	N/A
2004/0171998	12/2003	Marasco, Jr.	N/A	N/A
2004/0193218	12/2003	Butler	N/A	N/A
2004/0200094	12/2003	Baychar	N/A	N/A
2004/0241214	12/2003	Kirkwood et al.	N/A	N/A
2004/0243042	12/2003	Lipman	N/A	N/A
2005/0015036	12/2004	Lutri et al.	N/A	N/A
2005/0020955	12/2004	Sanders et al.	N/A	N/A
2005/0131327	12/2004	Lockwood et al.	N/A	N/A

2005/0230422	12/2004	Muller et al.	N/A	N/A
2005/0234417	12/2004	Yoshimasa et al.	N/A	N/A
2006/0009744	12/2005	Erdman et al.	N/A	N/A
2006/0020234	12/2005	Chou et al.	N/A	N/A
2006/0029650	12/2005	Coffey	N/A	N/A
2006/0070458	12/2005	Jones et al.	N/A	N/A
2006/0079599	12/2005	Arthur	N/A	N/A
2006/0100586	12/2005	Karpowicz et al.	N/A	N/A
2006/0100597	12/2005	Miskie	N/A	N/A
2006/0122548	12/2005	Abrams	N/A	N/A
2006/0161123	12/2005	Kudo et al.	N/A	N/A
2006/0173514	12/2005	Biel et al.	N/A	N/A
2006/0206047	12/2005	Lampe et al.	N/A	N/A
2006/0253082	12/2005	McIntosh et al.	N/A	N/A
2006/0273109	12/2005	Keller	N/A	N/A
2006/0282028	12/2005	Howard et al.	N/A	N/A
2007/0003604	12/2006	Jones	N/A	N/A
2007/0004896	12/2006	Ito et al.	N/A	N/A
2007/0009580	12/2006	DiCosmo et al.	N/A	N/A
2007/0027414	12/2006	Hoffman et al.	N/A	N/A
2007/0032762	12/2006	Vogel	N/A	N/A
2007/0032763	12/2006	Vogel	N/A	N/A
2007/0038172	12/2006	Zamierowski	N/A	N/A
2007/0040454	12/2006	Freudenberger et al.	N/A	N/A
2007/0055209	12/2006	Patel et al.	N/A	N/A
2007/0078467	12/2006	Mullen	N/A	N/A
2007/0100308	12/2006	Miyairi	N/A	N/A
2007/0141101	12/2006	Nugent et al.	N/A	N/A
2007/0147947	12/2006	Stenton et al.	N/A	N/A
2007/0164047	12/2006	Reidt et al.	N/A	N/A
2007/0185463	12/2006	Mulligan	N/A	N/A
2007/0203062	12/2006	Ellis-Behnke et al.	N/A	N/A
2007/0225663	12/2006	Watt et al.	N/A	N/A
2007/0239232	12/2006	Kurtz et al.	N/A	N/A
2007/0260207	12/2006	Ugander et al.	N/A	N/A
2007/0282236	12/2006	LaGreca	N/A	N/A
2008/0004549	12/2007	Anderson et al.	N/A	N/A
2008/0031748	12/2007	Ihle et al.	N/A	N/A
2008/0058691	12/2007	Sorensen	N/A	N/A
2008/0060550	12/2007	MacDonald et al.	N/A	N/A
2008/0089173	12/2007	Lu et al.	N/A	N/A
2008/0132821	12/2007	Propp et al.	N/A	N/A
2008/0167592	12/2007	Greer	N/A	N/A
2008/0167593	12/2007	Fleischmann	604/304	A61M 1/964
2008/0200905	12/2007	Heaton et al.	N/A	N/A
2008/0206155	12/2007	Tamarkin et al.	N/A	N/A
2008/0208147	12/2007	Argenta et al.	N/A	N/A
2008/0208163	12/2007	Wilkie	N/A	N/A
2008/0223378	12/2007	Henderson et al.	N/A	N/A
2008/0254103	12/2007	Harris et al.	N/A	N/A
2008/0287880	12/2007	Keller	N/A	N/A
2008/0294147	12/2007	Radl et al.	N/A	N/A
2008/0300555	12/2007	Olson et al.	N/A	N/A
2008/0306407	12/2007	Taylor	N/A	N/A
2008/0306456	12/2007	Riesinger	N/A	N/A

2008/0312572	12/2007	Riesinger	N/A	N/A
2008/0314929	12/2007	Keller	N/A	N/A
2009/0005746	12/2008	Nielsen et al.	N/A	N/A
2009/0012441	12/2008	Mulligan	N/A	N/A
2009/0020561	12/2008	Keller	N/A	N/A
2009/0022779	12/2008	Kelly et al.	N/A	N/A
2009/0030086	12/2008	Eady et al.	N/A	N/A
2009/0036873	12/2008	Nielsen et al.	N/A	N/A
2009/0054855	12/2008	Blott et al.	N/A	N/A
2009/0082740	12/2008	Lockwood et al.	N/A	N/A
2009/0093550	12/2008	Rolfes et al.	N/A	N/A
2009/0098073	12/2008	MacDonald et al.	N/A	N/A
2009/0125004	12/2008	Shen et al.	N/A	N/A
2009/0131892	12/2008	Karpowicz et al.	N/A	N/A
2009/0134186	12/2008	Keller	N/A	N/A
2009/0137973	12/2008	Karpowicz et al.	N/A	N/A
2009/0157017	12/2008	Ambrosio	N/A	N/A
2009/0157024	12/2008	Song	N/A	N/A
2009/0196844	12/2008	Choi et al.	N/A	N/A
2009/0198201	12/2008	Adahan	N/A	N/A
2009/0204085	12/2008	Biggie et al.	N/A	N/A
2009/0216170	12/2008	Robinson et al.	N/A	N/A
2009/0221977	12/2008	Blott et al.	N/A	N/A
2009/0221990	12/2008	Jaeb et al.	N/A	N/A
2009/0227935	12/2008	Zanella et al.	N/A	N/A
2009/0234306	12/2008	Vitaris	N/A	N/A
2009/0254053	12/2008	Svensby et al.	N/A	N/A
2009/0270820	12/2008	Johnson et al.	N/A	N/A
2009/0275872	12/2008	Addison et al.	N/A	N/A
2009/0275922	12/2008	Coulthard et al.	N/A	N/A
2009/0293887	12/2008	Wilkes et al.	N/A	N/A
2009/0299249	12/2008	Wilkes et al.	N/A	N/A
2009/0299251	12/2008	Buan	N/A	N/A
2009/0299255	12/2008	Kazala, Jr. et al.	N/A	N/A
2009/0299257	12/2008	Long et al.	N/A	N/A
2009/0299306	12/2008	Buan	N/A	N/A
2009/0299308	12/2008	Kazala, Jr. et al.	N/A	N/A
2009/0299340	12/2008	Kazala, Jr. et al.	N/A	N/A
2009/0312723	12/2008	Blott et al.	N/A	N/A
2010/0004611	12/2009	Aali	N/A	N/A
2010/0010462	12/2009	Kurata	N/A	N/A
2010/0016767	12/2009	Jones et al.	N/A	N/A
2010/0022972	12/2009	Lina et al.	N/A	N/A
2010/0030170	12/2009	Keller et al.	N/A	N/A
2010/0030171	12/2009	Canada et al.	N/A	N/A
2010/0030178	12/2009	Macmeccan et al.	N/A	N/A
2010/0036305	12/2009	Green	N/A	N/A
2010/0036334	12/2009	Heagle et al.	N/A	N/A
2010/0055158	12/2009	Vitaris et al.	N/A	N/A
2010/0069858	12/2009	Olson	N/A	N/A
2010/0069863	12/2009	Olson	N/A	N/A
2010/0084074	12/2009	McClernon et al.	N/A	N/A
2010/0106112	12/2009	Vogel	N/A	N/A
2010/0106120	12/2009	Holm	N/A	N/A
2010/0106121	12/2009	Holm	N/A	N/A

2010/0121286	12/2009	Locke et al.	N/A	N/A
2010/0122417	12/2009	Vrzalik et al.	N/A	N/A
2010/0125234	12/2009	Smith	N/A	N/A
2010/0125258	12/2009	Coulthard et al.	N/A	N/A
2010/0150991	12/2009	Bernstein	N/A	N/A
2010/0159192	12/2009	Cotton	N/A	N/A
2010/0168633	12/2009	Bougherara et al.	N/A	N/A
2010/0191196	12/2009	Heagle	N/A	N/A
2010/0191198	12/2009	Heagle	N/A	N/A
2010/0196106	12/2009	Allen	N/A	N/A
2010/0210986	12/2009	Sanders et al.	N/A	N/A
2010/0217177	12/2009	Cali et al.	N/A	N/A
2010/0230467	12/2009	Criscuolo et al.	N/A	N/A
2010/0249733	12/2009	Blott et al.	N/A	N/A
2010/0259406	12/2009	Caso et al.	N/A	N/A
2010/0262090	12/2009	Riesinger	N/A	N/A
2010/0262091	12/2009	Larsson	N/A	N/A
2010/0280468	12/2009	Haggstrom et al.	N/A	N/A
2010/0305490	12/2009	Coulthard et al.	N/A	N/A
2010/0318043	12/2009	Malhi et al.	N/A	N/A
2010/0318052	12/2009	Ha et al.	N/A	N/A
2011/0004172	12/2010	Eckstein et al.	N/A	N/A
2011/0015593	12/2010	Svedman et al.	N/A	N/A
2011/0021431	12/2010	Jones et al.	N/A	N/A
2011/0021964	12/2010	Larsen et al.	N/A	N/A
2011/0028919	12/2010	Johnnison et al.	N/A	N/A
2011/0028921	12/2010	Hartwell et al.	N/A	N/A
2011/0033503	12/2010	Sinko et al.	N/A	N/A
2011/0034888	12/2010	Aali	N/A	N/A
2011/0034892	12/2010	Buan	N/A	N/A
2011/0052664	12/2010	Tennican et al.	N/A	N/A
2011/0054422	12/2010	Locke et al.	N/A	N/A
2011/0054423	12/2010	Blott et al.	N/A	N/A
2011/0086077	12/2010	McCrea et al.	N/A	N/A
2011/0092927	12/2010	Wilkes et al.	N/A	N/A
2011/0092958	12/2010	Jacobs	N/A	N/A
2011/0098621	12/2010	Fabo et al.	N/A	N/A
2011/0105963	12/2010	Hu et al.	N/A	N/A
2011/0106030	12/2010	Scholz	N/A	N/A
2011/0112492	12/2010	Bharti et al.	N/A	N/A
2011/0118683	12/2010	Weston	N/A	N/A
2011/0125066	12/2010	Robinson et al.	N/A	N/A
2011/0130712	12/2010	Topaz	N/A	N/A
2011/0137222	12/2010	Masini	N/A	N/A
2011/0172617	12/2010	Riesinger	N/A	N/A
2011/0178375	12/2010	Forster	N/A	N/A
2011/0218509	12/2010	Dontas	N/A	N/A
2011/0224631	12/2010	Simmons et al.	N/A	N/A
2011/0245682	12/2010	Robinson et al.	N/A	N/A
2011/0245788	12/2010	Marquez	N/A	N/A
2011/0247636	12/2010	Pollack	N/A	N/A
2011/0257572	12/2010	Locke et al.	N/A	N/A
2011/0295220	12/2010	Heaton et al.	N/A	N/A
2012/0041402	12/2011	Greener	N/A	N/A
2012/0051945	12/2011	Orndorff et al.	N/A	N/A

2012/0065664	12/2011	Avitable et al.	N/A	N/A
2012/0095380	12/2011	Gergely et al.	N/A	N/A
2012/0095426	12/2011	Visscher et al.	N/A	N/A
2012/0101465	12/2011	McGuire, Jr.	N/A	N/A
2012/0123311	12/2011	Weidemann-Hendrickson et al.	N/A	N/A
2012/0123358	12/2011	Hall et al.	N/A	N/A
2012/0172778	12/2011	Rastegar et al.	N/A	N/A
2012/0203145	12/2011	Nilsson	N/A	N/A
2012/0203189	12/2011	Barta et al.	N/A	N/A
2012/0222687	12/2011	Czajka, Jr. et al.	N/A	N/A
2012/0232502	12/2011	Lowing	N/A	N/A
2012/0238932	12/2011	Atteia et al.	N/A	N/A
2012/0253302	12/2011	Corley	N/A	N/A
2013/0090615	12/2012	Jaeb et al.	N/A	N/A
2013/0090616	12/2012	Neubauer	N/A	N/A
2013/0102979	12/2012	Coulthard et al.	N/A	N/A
2013/0138054	12/2012	Fleischmann	N/A	N/A
2013/0150814	12/2012	Buan	N/A	N/A
2013/0165878	12/2012	Heagle	N/A	N/A
2013/0274688	12/2012	Weston	N/A	N/A
2013/0296762	12/2012	Toth	N/A	N/A
2013/0302545	12/2012	Schnelker et al.	N/A	N/A
2013/0338614	12/2012	Heaton et al.	N/A	N/A
2014/0114268	12/2013	Auguste et al.	N/A	N/A
2014/0155849	12/2013	Heaton et al.	N/A	N/A
2014/0228791	12/2013	Hartwell	N/A	N/A
2014/0236109	12/2013	Greener	N/A	N/A
2014/0309601	12/2013	Hall et al.	N/A	N/A
2015/0065965	12/2014	Haggstrom et al.	N/A	N/A
2015/0209492	12/2014	Blott et al.	N/A	N/A
2015/0320604	12/2014	Adie et al.	N/A	N/A
2016/0051737	12/2015	Joshi et al.	N/A	N/A
2016/0081859	12/2015	Hartwell	N/A	N/A
2016/0317357	12/2015	Vitaris et al.	N/A	N/A
2016/0354086	12/2015	Dunn	N/A	N/A
2017/0095598	12/2016	Joshi et al.	N/A	N/A
2017/0128642	12/2016	Buan	N/A	N/A
2017/0181896	12/2016	Hartwell	N/A	N/A
2017/0181897	12/2016	Hartwell	N/A	N/A
2017/0312407	12/2016	Coulthard et al.	N/A	N/A
2018/0147091	12/2017	Greener et al.	N/A	N/A
2018/0369462	12/2017	Anderson et al.	N/A	N/A
2019/0008690	12/2018	Adie et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2015201022	12/2015	AU	N/A
2705898	12/2008	CA	N/A
2653146	12/2008	CA	N/A
2739243	12/2009	CA	N/A
101385887	12/2008	CN	N/A
101404965	12/2008	CN	N/A
847475	12/1951	DE	N/A
3137839	12/1982	DE	N/A

3032092	12/1983	DE	N/A
3443101	12/1985	DE	N/A
3935818	12/1990	DE	N/A
4012232	12/1990	DE	N/A
9017289	12/1991	DE	N/A
19700271	12/1996	DE	N/A
19832634	12/1999	DE	B32B 7/022
19844355	12/1999	DE	N/A
202004017052	12/2004	DE	N/A
0091800	12/1982	EP	N/A
0020662	12/1983	EP	N/A
0122042	12/1983	EP	N/A
0257916	12/1987	EP	N/A
0340018	12/1988	EP	N/A
0355186	12/1989	EP	N/A
0507459	12/1991	EP	N/A
0521434	12/1992	EP	N/A
0541251	12/1992	EP	N/A
0617152	12/1993	EP	D04B 1/16
0619105	12/1993	EP	N/A
0752839	12/1997	EP	A61F 5/01
0858810	12/1997	EP	N/A
0777504	12/1997	EP	N/A
0888141	12/1998	EP	N/A
0782421	12/1998	EP	N/A
0941726	12/1998	EP	N/A
1007015	12/1999	EP	N/A
1013290	12/1999	EP	N/A
1029585	12/1999	EP	N/A
1068451	12/2000	EP	N/A
1105171	12/2000	EP	N/A
1105180	12/2000	EP	N/A
1107813	12/2000	EP	N/A
1030657	12/2000	EP	N/A
1169071	12/2001	EP	N/A
1283702	12/2002	EP	N/A
0708620	12/2002	EP	N/A
1306123	12/2002	EP	N/A
1088569	12/2002	EP	N/A
1411874	12/2003	EP	N/A
1440737	12/2003	EP	N/A
1018967	12/2003	EP	N/A
0999858	12/2003	EP	N/A
1452156	12/2003	EP	N/A
1565219	12/2004	EP	N/A
1440667	12/2005	EP	N/A
1637088	12/2005	EP	N/A
1284777	12/2005	EP	N/A
1263366	12/2005	EP	N/A
0982015	12/2005	EP	N/A
1448261	12/2006	EP	N/A
1171065	12/2006	EP	N/A
1767177	12/2006	EP	N/A
1807032	12/2006	EP	N/A
1880840	12/2007	EP	N/A

1476217	12/2007	EP	N/A
1897569	12/2007	EP	N/A
1904137	12/2007	EP	N/A
1922045	12/2007	EP	N/A
1923077	12/2007	EP	N/A
1931413	12/2007	EP	N/A
1955887	12/2007	EP	N/A
1985270	12/2007	EP	N/A
1121163	12/2007	EP	N/A
2059204	12/2008	EP	N/A
2079507	12/2008	EP	N/A
2098257	12/2008	EP	N/A
2106255	12/2008	EP	N/A
2109472	12/2008	EP	N/A
2111804	12/2008	EP	N/A
2140968	12/2009	EP	N/A
2161011	12/2009	EP	N/A
2172164	12/2009	EP	N/A
2185206	12/2009	EP	N/A
1807031	12/2009	EP	N/A
2203137	12/2009	EP	N/A
2244746	12/2009	EP	N/A
2253353	12/2009	EP	N/A
2254537	12/2009	EP	N/A
2004116	12/2010	EP	N/A
2326295	12/2010	EP	N/A
2335749	12/2010	EP	N/A
2349155	12/2010	EP	N/A
1578477	12/2010	EP	N/A
2420214	12/2011	EP	N/A
2021046	12/2011	EP	N/A
2178573	12/2011	EP	N/A
2462908	12/2011	EP	N/A
2021047	12/2012	EP	N/A
2711034	12/2013	EP	N/A
2305325	12/2013	EP	N/A
2345437	12/2013	EP	N/A
1339366	12/2013	EP	N/A
2544642	12/2014	EP	N/A
2648668	12/2014	EP	N/A
2931197	12/2014	EP	N/A
2687245	12/2015	EP	N/A
3085344	12/2015	EP	N/A
3139878	12/2016	EP	N/A
1163907	12/1957	FR	N/A
114754	12/1917	GB	N/A
236350	12/1924	GB	N/A
641061	12/1949	GB	N/A
821959	12/1958	GB	N/A
1224009	12/1970	GB	N/A
1255395	12/1970	GB	N/A
1400124	12/1974	GB	N/A
1549756	12/1978	GB	N/A
2021241	12/1978	GB	N/A
2093190	12/1981	GB	N/A

2099306	12/1981	GB	N/A
2195255	12/1987	GB	N/A
2288734	12/1994	GB	N/A
2307180	12/1996	GB	N/A
2378392	12/2002	GB	N/A
2415908	12/2005	GB	N/A
2424582	12/2005	GB	N/A
2435419	12/2006	GB	N/A
2435422	12/2006	GB	N/A
2468905	12/2009	GB	N/A
S5987824	12/1983	JP	N/A
S61288860	12/1985	JP	N/A
H02139624	12/1989	JP	N/A
H09505752	12/1996	JP	N/A
2001011759	12/2000	JP	N/A
2004521665	12/2001	JP	N/A
2003165843	12/2002	JP	N/A
2003250879	12/2002	JP	N/A
2003532504	12/2002	JP	N/A
2004509658	12/2003	JP	N/A
2009506878	12/2008	JP	N/A
2009509570	12/2008	JP	N/A
2009536852	12/2008	JP	N/A
2010504816	12/2009	JP	N/A
2010516387	12/2009	JP	N/A
2012024124	12/2011	JP	N/A
2014518651	12/2013	JP	N/A
1251912	12/1985	SU	N/A
WO-8300742	12/1982	WO	N/A
WO-8401904	12/1983	WO	N/A
WO-9011795	12/1989	WO	N/A
WO-9100718	12/1990	WO	N/A
WO-9209301	12/1991	WO	N/A
WO-9209651	12/1991	WO	N/A
WO-9216245	12/1991	WO	N/A
WO-9220299	12/1991	WO	N/A
WO-9306802	12/1992	WO	N/A
WO-9309176	12/1992	WO	N/A
WO-9420133	12/1993	WO	N/A
WO-9423677	12/1993	WO	N/A
WO-9504511	12/1994	WO	N/A
WO-9514451	12/1994	WO	N/A
WO-9516416	12/1994	WO	N/A
WO-9529959	12/1994	WO	N/A
WO-9605873	12/1995	WO	N/A
WO-9621410	12/1995	WO	N/A
WO-9640174	12/1995	WO	N/A
WO-9703717	12/1996	WO	N/A
WO-9711658	12/1996	WO	N/A
WO-9733922	12/1996	WO	N/A
WO-9742986	12/1996	WO	N/A
WO-9803267	12/1997	WO	N/A
WO-9806444	12/1997	WO	N/A
WO-9901173	12/1998	WO	N/A
WO-9917698	12/1998	WO	N/A

WO-9930629	12/1998	WO	N/A
WO-9939671	12/1998	WO	N/A
WO-9947097	12/1998	WO	N/A
WO-9965536	12/1998	WO	N/A
WO-0007653	12/1999	WO	N/A
WO-0038752	12/1999	WO	N/A
WO-0042957	12/1999	WO	N/A
WO-0050143	12/1999	WO	N/A
WO-0059424	12/1999	WO	N/A
WO-0061206	12/1999	WO	N/A
WO-0062827	12/1999	WO	N/A
WO-0064396	12/1999	WO	N/A
WO-0119430	12/2000	WO	N/A
WO-0134223	12/2000	WO	N/A
WO-0162312	12/2000	WO	N/A
WO-0166017	12/2000	WO	N/A
WO-0185248	12/2000	WO	N/A
WO-0202079	12/2001	WO	N/A
WO-0217840	12/2001	WO	N/A
WO-0226180	12/2001	WO	N/A
WO-0238096	12/2001	WO	N/A
WO-02076379	12/2001	WO	N/A
WO-02083046	12/2001	WO	N/A
WO-02092783	12/2001	WO	N/A
WO-02094256	12/2001	WO	N/A
WO-02102864	12/2001	WO	N/A
WO-03045492	12/2002	WO	N/A
WO-03057307	12/2002	WO	N/A
WO-03092620	12/2002	WO	N/A
WO-2004024300	12/2003	WO	N/A
WO-2004037334	12/2003	WO	N/A
WO-2004054632	12/2003	WO	N/A
WO-2004073566	12/2003	WO	N/A
WO-2004077387	12/2003	WO	N/A
WO-2005009488	12/2004	WO	N/A
WO-2005016179	12/2004	WO	N/A
WO-2005017000	12/2004	WO	N/A
WO-2005018695	12/2004	WO	N/A
WO-2005025447	12/2004	WO	N/A
WO-2005025666	12/2004	WO	N/A
WO-2005046761	12/2004	WO	N/A
WO-2005051461	12/2004	WO	N/A
WO-2005070480	12/2004	WO	N/A
WO-2005082435	12/2004	WO	N/A
WO-2005105180	12/2004	WO	N/A
WO-2005118011	12/2004	WO	N/A
WO-2005123170	12/2004	WO	N/A
WO-2006014534	12/2005	WO	N/A
WO-2006030054	12/2005	WO	N/A
WO-2006034128	12/2005	WO	N/A
WO-2006048246	12/2005	WO	N/A
WO-2006052745	12/2005	WO	N/A
WO-2006052839	12/2005	WO	N/A
WO-2006056294	12/2005	WO	N/A
WO-2006081403	12/2005	WO	N/A

WO-2006087021	12/2005	WO	N/A
WO-2006116992	12/2005	WO	N/A
WO-2006135506	12/2005	WO	N/A
WO-2007002835	12/2006	WO	N/A
WO-2007013064	12/2006	WO	N/A
WO-2007019038	12/2006	WO	N/A
WO-2007024230	12/2006	WO	N/A
WO-2007030598	12/2006	WO	N/A
WO-2007030601	12/2006	WO	N/A
WO-2007059742	12/2006	WO	N/A
WO-2007082538	12/2006	WO	N/A
WO-2007084792	12/2006	WO	N/A
WO-2007085396	12/2006	WO	N/A
WO-2007092397	12/2006	WO	N/A
WO-2007106592	12/2006	WO	N/A
WO-2007106594	12/2006	WO	N/A
WO-2007124198	12/2006	WO	N/A
WO-2007133618	12/2006	WO	N/A
WO-2008013896	12/2007	WO	N/A
WO-2008027449	12/2007	WO	N/A
WO-2008036225	12/2007	WO	N/A
WO-2008039223	12/2007	WO	N/A
WO-2008039314	12/2007	WO	N/A
WO-2008039839	12/2007	WO	N/A
WO-2008040020	12/2007	WO	N/A
WO-2008041926	12/2007	WO	N/A
WO-2008049277	12/2007	WO	N/A
WO-2008064502	12/2007	WO	N/A
WO-2008076407	12/2007	WO	N/A
WO-2008082444	12/2007	WO	N/A
WO-2008091251	12/2007	WO	N/A
WO-2008100437	12/2007	WO	N/A
WO-2008100440	12/2007	WO	N/A
WO-2008104609	12/2007	WO	N/A
WO-2008112304	12/2007	WO	N/A
WO-2008131895	12/2007	WO	N/A
WO-2008134544	12/2007	WO	N/A
WO-2008134774	12/2007	WO	N/A
WO-2009002260	12/2007	WO	N/A
WO-2009019227	12/2008	WO	N/A
WO-2009019229	12/2008	WO	N/A
WO-2009021523	12/2008	WO	N/A
WO-2009042514	12/2008	WO	N/A
WO-2009047524	12/2008	WO	N/A
WO-2009052193	12/2008	WO	N/A
WO-2009060097	12/2008	WO	N/A
WO-2009060327	12/2008	WO	N/A
WO-2009062327	12/2008	WO	N/A
WO-2009066104	12/2008	WO	N/A
WO-2009066105	12/2008	WO	N/A
WO-2009066106	12/2008	WO	N/A
WO-2009071932	12/2008	WO	N/A
WO-2009089390	12/2008	WO	N/A
WO-2009097534	12/2008	WO	N/A
WO-2009103031	12/2008	WO	N/A

WO-2009111655	12/2008	WO	N/A
WO-2009111657	12/2008	WO	N/A
WO-2009114790	12/2008	WO	N/A
WO-2009117635	12/2008	WO	N/A
WO-2009122989	12/2008	WO	N/A
WO-2009124100	12/2008	WO	N/A
WO-2009124125	12/2008	WO	N/A
WO-2009126102	12/2008	WO	N/A
WO-2009126103	12/2008	WO	N/A
WO-2009126833	12/2008	WO	N/A
WO-2009145703	12/2008	WO	N/A
WO-2009146441	12/2008	WO	N/A
WO-2009147402	12/2008	WO	N/A
WO-2009151380	12/2008	WO	N/A
WO-2009156709	12/2008	WO	N/A
WO-2009158124	12/2008	WO	N/A
WO-2009158127	12/2008	WO	N/A
WO-2009158128	12/2008	WO	N/A
WO-2009158130	12/2008	WO	N/A
WO-2010008167	12/2009	WO	N/A
WO-2010026251	12/2009	WO	N/A
WO-2010033574	12/2009	WO	N/A
WO-2010033613	12/2009	WO	N/A
WO-2010035017	12/2009	WO	N/A
WO-2010051418	12/2009	WO	N/A
WO-2010059849	12/2009	WO	N/A
WO-2010072309	12/2009	WO	N/A
WO-2010072395	12/2009	WO	N/A
WO-2010082872	12/2009	WO	N/A
WO-2010089448	12/2009	WO	N/A
WO-2010120776	12/2009	WO	N/A
WO-2010139926	12/2009	WO	N/A
WO-2010141271	12/2009	WO	N/A
WO-2010142959	12/2009	WO	N/A
WO-2010147533	12/2009	WO	N/A
WO-2010147592	12/2009	WO	N/A
WO-2011019476	12/2010	WO	N/A
WO-2011023275	12/2010	WO	N/A
WO-2011023650	12/2010	WO	N/A
WO-2011063818	12/2010	WO	N/A
WO-2011115908	12/2010	WO	N/A
WO-2011128651	12/2010	WO	N/A
WO-2011135284	12/2010	WO	N/A
WO-2011135285	12/2010	WO	N/A
WO-2011135286	12/2010	WO	N/A
WO-2011135287	12/2010	WO	N/A
WO-2011144888	12/2010	WO	N/A
WO-2012009370	12/2011	WO	N/A
WO-2012022484	12/2011	WO	N/A
WO-2012028842	12/2011	WO	N/A
WO-2012041296	12/2011	WO	N/A
WO-2012074512	12/2011	WO	N/A
WO-2012131237	12/2011	WO	N/A
WO-2012140378	12/2011	WO	N/A
WO-2012143665	12/2011	WO	N/A

WO-2012146656	12/2011	WO	N/A
WO-2012150235	12/2011	WO	N/A
WO-2015172108	12/2014	WO	N/A

OTHER PUBLICATIONS

Brief communication—Letter from the Opponent of Feb. 24, 2023 for European Patent No. 3628289, mailed on Mar. 1, 2023, 11 pages. cited by applicant

Brief communication—Letter from the Proprietor of the Patent of Mar. 17, 2023 for European Patent No. 3628289, mailed on Mar. 23, 2023, 5 pages. cited by applicant

Board of Appeal—Letter of the Patent Proprietor dated May 22, 2023 for European Patent No. 2821035, mailed on May 26, 2023, 10 pages. cited by applicant

Achterberg V., et al., "Hydroactive Dressings and Serum Proteins: An In Vitro Study," Journal of Wound Care, vol. 5 (2), Feb. 1996, pp. 79-82. cited by applicant

Advantec MFS, Inc., "Membrane Filters" (catalog), retrieved from http://www.advantecmfs.com/catalog/filt/membrane.pdf, on Jan. 29, 2016, Copyright 2001-2011, 17 pages. cited by applicant

Annex to the Communication, the Opposition of European Patent No. 2563306, mailed on Oct. 6, 2017, 15 pages. cited by applicant

Argenta L C., et al., "Vacuum-Assisted Closure: A New Method for Wound Control and Treatment: Clinical Experience," Annals of Plastic Surgery, vol. 38 (6), Jun. 1997, pp. 563-577. cited by applicant

Arnljots B., et al., "Irrigation Treatment in Split-Thickness Skin Grafting of Intractable Leg Ulcers," Scandinavian Journal of Plastic and Reconstructive Surgery, vol. 19, 1985, pp. 211-213. cited by applicant

Aubrey D.A., et al., "Treatment of the Perineal Wound after Proctectomy by Intermittent Irrigation," Arch. Surg, vol. 119, Oct. 1984, pp. 1141-1144. cited by applicant

Bagautdinov N.A., "Variant of External Vacuum Aspiration in the Treatment of Purulent Diseases of Soft Tissues," in Current Problems in Modern Clinical Surgery, Interdepartmental Collection, 1986, pp. 94-96. cited by applicant

Bevan D., et al., "Diverse and potent activities of HGF/SF in skin wound repair," Journal of Pathology, vol. 203, 2004, pp. 831-838. cited by applicant

Brief Communication of Letter from the proprietor of the patent and copies of Auxiliary requests 1-2 for European Patent No. 2821035 mailed on Feb. 12, 2021, 27 pages. cited by applicant

Chariker M.E., et al., "Effective Management of Incisional and Cutaneous Fistulae with Closed Suction Wound Drainage," Contemporary Surgery, vol. 34, Jun. 1989, pp. 59-63. cited by applicant

Chintamani et al., "Half Versus Full Vacuum Suction Drainage After Modified Radical Mastectomy for Breast Cancer—a Prospective Randomized Clinical Trial," BMC Cancer, Research Article, vol. 5(11), Jan. 27, 2005, 5 pages. cited by applicant

Communication of Notice of Opposition dated Dec. 17, 2015, and Opposition of European patent EP2563308 B1, dated Dec. 11, 2015, on behalf of KCI Licensing, Inc., and cited publications D1-D6, 250 pages. cited by applicant

Communication of the Board of Appeal to Patent Proprietor for European Patent No. 2563308, mailed on Jul. 2, 2021, 6 pages. cited by applicant

Data Sheet for the decision for Jul. 17, 2020 cited during the appeal procedure for European Patent No. 2563308, mailed on Jan. 22, 2021, 37 pages. cited by applicant

Davydov Y A., et al., "Concepts for Clinical Biological Management of the Wound Process in the Treatment of Purulent Wounds Using Vacuum Therapy," Vestnik Khirugii, Feb. 1991, pp. 15-17. cited by applicant

Davydov Y.A., et al., "The Bacteriological and Cytological Assessment of Vacuum Therapy of Purulent Wounds," Vestnik Khirurgii, Oct. 1988, pp. 11-14. cited by applicant

Davydov Y.A., et al., "Vacuum Therapy in the Treatment of Purulent Lactation Mastitis," The Kremlin Papers: Perspectives in Wound Care, Vestnik Khirurgii, BlueSky Publishing, 2004, pp. 5-7. cited by applicant

De Lange M.Y., et al., "Vacuum-Assisted Closure: Indications and Clinical Experience," Eur J Plast Surg (2000), vol. 23, Feb. 9, 2000, pp. 178-182. cited by applicant

Declaration of Chris Locke in support of the opposition to European Patent No. 2563308 dated Sep. 7, 2016, 1 page. cited by applicant

Declaration of Chris Locke Submitted in the Opposition against European Patent No. 2231221, mailed on Sep. 7, 2016, 1 page. cited by applicant

Declaration of Chris Locke submitted in the Opposition against European Patent No. EP2563308, dated Sep. 7, 2016, 6 pages. cited by applicant

Dilmaghani A., et al., “A Method for Closed Irrigation and Suction Therapy in Deep Wound Infections,” Journal of Bone and Joint Surgery, Mar. 1969, vol. 51-A(2), pp. 323-342. cited by applicant
Grounds for the Decision, the Opposition of European Patent No. 2563306, mailed on Sep. 7, 2018, 101 pages. cited by applicant
Hartz R.S., et al., “Healing of the Perineal Wound,” The Archives of Surgery, Apr. 1980, vol. 115, pp. 471-474. cited by applicant
Health Technology Literature Review, “Vacuum Assisted Closure Therapy for Wound Care,” The Medical Advisory Secretariat, Dec. 2004, pp. 1-57. cited by applicant
Hersle K., et al., “Uses of Dextranomer Absorbent Pads After Cryosurgery of Cutaneous Malignancies,” The Journal of Dermatologic Surgery and Oncology, vol. 8, Jan. 1982, pp. 35-37. cited by applicant
Information about the Result of Oral Proceedings for the Opposition of European Patent No. EP2563306, dated Jun. 21, 2018, 1 page. cited by applicant
Information about the Result of Oral Proceedings for the Opposition of European Patent No. EP2563308, dated Jul. 21, 2017, 3 pages. cited by applicant
Interlocutory Decision and Annex to the Communication for the Opposition of European Patent No. EP2563308, dated Aug. 14, 2017, 76 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/GB2011/000622, mailed on Nov. 8, 2012, 7 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/GB2011/000625, mailed on Nov. 8, 2012, 8 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/GB2011/000626, mailed on Nov. 8, 2012, 8 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/GB2011/000628, mailed on Nov. 8, 2012, 8 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/GB2011/000629, mailed on Oct. 31, 2013, 8 pages. cited by applicant
International Preliminary Report on Patentability for Application No. PCT/GB2011/00621, mailed on Nov. 29, 2012, 7 pages. cited by applicant
International Search Report for Application No. PCT/GB2008/051088, mailed on Mar. 4, 2009, 2 pages. cited by applicant
International Search Report for Application No. PCT/GB2008/051089, mailed on Mar. 4, 2009, 3 pages. cited by applicant
International Search Report for Application No. PCT/GB2008/051090, mailed on Mar. 4, 2009, 2 pages. cited by applicant
International Search Report for Application No. PCT/GB2011/000621, mailed on Jul. 26, 2011, 4 pages. cited by applicant
International Search Report for Application No. PCT/GB2011/000622, mailed on Jul. 28, 2011, 4 pages. cited by applicant
International Search Report for Application No. PCT/GB2011/000625, mailed on Aug. 10, 2011, 5 pages. cited by applicant
International Search Report for Application No. PCT/GB2011/000626, mailed on Aug. 4, 2011, 5 pages. cited by applicant
International Search Report for Application No. PCT/GB2011/000628, mailed on Sep. 6, 2011, 7 pages. cited by applicant
International Search Report for Application No. PCT/GB2011/000629, mailed on Dec. 23, 2011, 6 pages. cited by applicant
Invalidity Suit by *KCI Medizinprodukte GmbH* versus *Kalypto Medical, Inc.*, concerning declaration of invalidity of the German part of the European Patent No. 2021046 (German application No. 602007021330.4) dated Mar. 11, 2015 in 66 pages. EP2021046 is related to the present application by virtue of a common priority claim to U.S. Appl. No. 11/432,855, now U.S. Pat. No. 7,615,036, and U.S. Appl. No. 11/610,458, now U.S. Pat. No. 7,779,625. cited by applicant
Kalypto Medical, “NPD 1000 Negative Pressure Wound Care System,” Clinician & Patient Instructions for Use, Feb. 2010, 36 pages. cited by applicant
Kalypto Medical, NPD 1000 Product Brochure (publication date unknown, believed to be Nov. 2010), 4 pages. cited by applicant

KCI Licensing Inc, “Prevena™ Incision Management System—Clinician Guide—Instructions for Use,” 390061 Rev D, Jan. 2010, 10 pages. cited by applicant

KCI, “Prevena—Incision Management System Patient Guide,” Jan. 2010, 2 pages. cited by applicant

Kendall ULTEC Hydrocolloid Dressing (4×4“),” Product Ordering Page, web page downloaded on Jul. 13, 2014, 1 page. cited by applicant

Khirugii V., “A Collection of Published Studies Complementing the Research and Innovation of Wound Care,” The Kremlin Papers, Perspectives in Wound Care, Russian Medical Journal, Blue Sky Publishing, 2004, pp. 2-17. cited by applicant

Kostiuchenok B.M., et al., “The Vacuum Effect in the Surgical Treatment of Purulent Wounds,” Russian Journal: Vestnik Khirurgii, Sep. 1986, pp. 18-21. cited by applicant

Letter of the opponent relating to oral proceedings during the appeal procedure for European Patent No. 2563308, mailed on Feb. 26, 2021, 3 pages .<gdiv class=“ginger-extension-definitionpopup” style=“left: 163.463px; top: 17.6px; z-index: 2147483646; display: none;”><gdiv class=“ginger-dp”> <gdiv class=“ginger-dp-content”> <gdiv class=“ginger-dp-title”><gspanid=“dp-title”>with</gspan></gdiv> <gdiv class=“ginger-dp-description” id=“dp-description”>accompanying</gdiv> <gdiv class=“ginger-dp-more”>More <gspan>(Definitions, Synonyms, Translation)</gspan></gdiv> .< /gdiv></gdiv></gdiv>. cited by applicant

Letter of the opponent relating to oral proceedings during the appeal procedure for European Patent No. 2563308, mailed on Sep. 6, 2021, 4 pages. cited by applicant

Letter of the Patent Proprietor dated Jan. 22, 2021 relating to the appeal procedure for European Patent No. 2563308, 9 pages. cited by applicant

Letter regarding the Opposition Procedure from the Opponent for the Opposition of European Patent No. 2563308, dated Nov. 10, 2016, 10 pages. cited by applicant

Letter relating to the Appeal Procedure for the Opposition of the European Patent No. 2563306, mailed on Jan. 17, 2019, 3 pages. cited by applicant

Letter relating to the Appeal Procedure for the Opposition of the European Patent No. 2563308, mailed on Apr. 17, 2018, in 6 pages. cited by applicant

Letter relating to the Appeal Procedure for the Opposition of the European Patent No. 2563308, mailed on Jun. 22, 2018, in 4 pages. cited by applicant

MacMillan Dictionary, “Comprise,” Retrieved from https://www.macmillandictionary.com/us/dictionary/american/comprise on Jan. 29, 2020, 4 pages. cited by applicant

Merriam-Webster, “Comprise,” Retrieved from https://www.merriam-webster.com/dictionary/comprise on Jan. 29, 2020, 8 pages. cited by applicant

Merriam-Webster, Definition of “Port”, retrieved from URL: https://www.merriam-webster.com/dictionary/port, on Oct. 26, 2020, 6 pages. cited by applicant

Minutes of the Oral Proceedings—Boards of Appeal, re the Opposition of European Patent No. 2563308, mailed on Nov. 4, 2021, 3 pages. cited by applicant

Mitchell R.N., et al., “Role of Stem Cells in Tissue Homeostasis,” Pocket Companion to Robbins and Cotran Pathologic Basis of Disease, 7th Edition, 2006, p. 55 (3 pages). cited by applicant

Morykwas M.J., et al., “Vacuum-Assisted Closure: A New Method for Wound Control and Treatment: Animal Studies and Basic Foundation,” Annals Plastic Surgery, vol. 38 (6), Jun. 1997, pp. 553-562. cited by applicant

Notice of Appeal by Appellant for the Opposition of the European Patent No. 2563308, mailed on Oct. 9, 2017, 4 pages. cited by applicant

Notice of Appeal by Proprietor for the Opposition of the European Patent No. 2563308, mailed on Oct. 10, 2017, 5 pages. cited by applicant

Notice of Opposition—Opponent's Statement of Facts and Arguments for the European Patent No. 2821035, mailed on Sep. 7, 2020, 12 pages. cited by applicant

Notice of Opposition—Statement of Facts and Evidence for the European Patent No. 2563306, mailed on Dec. 30, 2016, 19 pages. cited by applicant

Notice of Opposition—Statement of Facts and Evidence for the European Patent No. 2699276, mailed on May 3, 2017, 14 pages. cited by applicant

Notice of Oppositions from Opponent for the European Patent No. 2821035, mailed on Sep. 8, 2020, 73 pages. cited by applicant

Nursing75., “Wound Suction: Better Drainage with Fewer Problems,” vol. 5(10), Oct. 1975, pp. 52-55. cited by applicant

Office Action mailed Aug. 22, 2011 for Canadian Application No. 2739605, 2 pages. cited by applicant
Opponent's letter relating to the Appeal Procedure for the Opposition of the European Patent No. 2563306, mailed on Aug. 20, 2020, 37 pages. cited by applicant
Opponent's Written Submission in Preparation for the Oral Proceedings for the Opposition of European Patent No. 2563308, mailed on May 19, 2017, 10 pages. cited by applicant
Opponent's Written Submission in Preparation for the Oral Proceedings for the Opposition of European Patent No. EP2563306, dated Apr. 20, 2018, 14 pages. cited by applicant
Opponent's Written Submission in Preparation for the Oral Proceedings for the Opposition of European Patent No. EP2563306, dated Jun. 11, 2018, 7 pages. cited by applicant
Opponent's Written Submission in Preparation for the Oral Proceedings for the Opposition of European Patent No. EP2699276, dated Jul. 11, 2018, 3 pages. cited by applicant
Preparation for Oral Proceedings and Annex to the Communication for the Opposition of European Patent No. 2563308, mailed on Dec. 19, 2016, 18 pages. cited by applicant
Preparation for Oral Proceedings and Annex to the Communication for the Opposition of European Patent No. 2699276, mailed on Jan. 12, 2018, 19 pages. cited by applicant
Priority Document filed Apr. 27, 2010 for Great Britain Application No. 1006986.2, by Smith & Nephew PLC, 40 pages. cited by applicant
Proprietor's letter relating to the Appeal Procedure for the Opposition of the European Patent No. 2563306, mailed on Aug. 6, 2020, 32 pages. cited by applicant
Proprietor's Written Submission in Preparation for the Oral Proceedings, the Opposition of European Patent No. 2563306, mailed on Apr. 19, 2018, 39 pages. cited by applicant
Proprietor's Written Submission in Preparation for the Oral Proceedings, the Opposition of European Patent No. 2563308, mailed on Jul. 10, 2017, 12 pages. cited by applicant
Proprietor's Written Submission in Preparation for the Oral Proceedings, the Opposition of European Patent No. 2563308, mailed on May 17, 2017, 37 pages. cited by applicant
Proprietor's Written Submission in Preparation for the Oral Proceedings, the Opposition of European Patent No. 2699276, mailed on Jul. 10, 2018, 39 pages. cited by applicant
Protz K., “Modern Wound Dressings Support the Healing Process,” Wound care: Indications and Application, Geriatrie Journal, Apr. 2005, pp. 3333-3339 (17 pages with English translation). cited by applicant
Reply of the Patent Proprietor to the Notice of Opposition for the European Patent No. 2563306, mailed on Jun. 5, 2017, 44 pages. cited by applicant
Reply of the Patent Proprietor to the Notice of Opposition for the European Patent No. 2563308, mailed on Aug. 15, 2016, 35 pages. cited by applicant
Reply of the Patent Proprietor to the Notice of Opposition for the European Patent No. 2699276, mailed on Oct. 19, 2017, 5 pages. cited by applicant
Reply to Appeal—Letter of the patent proprietor for the European Patent No. 2563308, mailed on Aug. 17, 2021, 13 pages. cited by applicant
Reply to Appeal, in Opposition by KCI Licensing Inc. to Smith & Nephew PLC, the European Patent No. 2563308, mailed on May 24, 2018, 9 pages. cited by applicant
Reply to Grounds of Appeal, in Opposition by KCI Licensing Inc. to Smith & Nephew PLC for the European Patent No. 2563308, dated Apr. 27, 2018, 4 pages. cited by applicant
Reply to the Appeal mailed on May 21, 2019 for the Opposition of European Patent No. 2563306, 18 pages. cited by applicant
Smith & Nephew, “PICO Single Use Negative Pressure Wound Therapy System,” Spiral Booklet, Mar. 2011, 7 pages. cited by applicant
Solovev V. A, et al., “The Method of Treatment of Immature External Fistulas in the Upper Gastrointestinal Tract-Guidelines,” USSR Ministry of Health, S. M. Kirov Gorky State Medical Institute, 1987, 19 pages. cited by applicant
Solovev V.A., “Treatment and Prevention of Suture Failures after Gastric Resection,” Dissertation Abstract, Gorky, 1988, 51 pages. cited by applicant
Specification and drawings filed in application No. EP14187266.3, file on Jan. 8, 2010, 46 pages. cited by applicant
Statement of Grounds of Appeal by Appellant-Opponent for the Opposition of European Patent No. 2563308, dated Dec. 12, 2017, 6 pages. cited by applicant
Statement of Grounds of Appeal by Appellant-Proprietor for the Opposition of European Patent No. 2563308, mailed on Dec. 5, 2017, 26 pages. cited by applicant

Statement of Grounds of Appeal for the Opposition of European Patent No. 2563306, mailed on Jan. 16, 2019, 90 pages. cited by applicant
Stewart J., “World Wide Wounds—Next Generation of Products for Wound Management,” Nov. 2002, http://www.worldwidewounds.com/2003/aprii/Stewart/Next-Generation-Products.html, 13 pages. cited by applicant
Summons to attend oral proceedings pursuant to Rule 115(1) EPC for Patent No. 2821035, mailed on Jul. 5, 2021, 18 pages. cited by applicant
Svedman P., “A Dressing Allowing Continuous Treatment of a Biosurface,” IRCS Medical Science: Biomedical Technology; Clinical Medicine; Surgery and Transplantation, vol. 7, 1979, p. 221. cited by applicant
Svedman P., et al., “A Dressing System Providing Fluid Supply and Suction Drainage Used for Continuous or Intermittent Irrigation,” Annals of Plastic Surgery, vol. 17 (2), Aug. 1986, 9 pages. cited by applicant
Svedman P., “Irrigation Treatment of Leg Ulcers,” The Lancet, Sep. 3, 1983, pp. 532-534. cited by applicant
Swift S., et al., “Quorum Sensing in Aeromonas Hydrophila and Aeromonas Salmoncida: Identification of LuxRI Homologs AhyRI and AsaRI and Their Cognate N-Acylhomoserine Lactone Signal Molecules,” Journal of Bacteriology, Sep. 1997, vol. 179, No. 17, pp. 5271-5281. cited by applicant
Technology Watch, May 1989, 1 page. cited by applicant
Teder H., et al., “Continuous Wound Irrigation in the Pig,” Journal of Investigative Surgery, 1990, vol. 3, pp. 399-407. cited by applicant
Termination of the Opposition Proceedings with Maintenance of European Patent No. 2699276, dated Nov. 29, 2018, 1 page. cited by applicant
The Free Dictionary, “Welding,” The American Heritage®, Fourth Edition, 2000, 2 pages. cited by applicant
The Wayback Machine, “Comfort advantages with AirX™,” retrieved from http://web.archive.org/web/20090121000205/http://www.airx.eu:80/content/view/2/3/lang,en/, on Jan. 21, 2009, 1 page. cited by applicant
The Wayback Machine, “Moisture-Transporting Material,” retrieved from http://web.archive.org/web/20090121001036/http://www.airx.eu/content/view/1/14/lang,en/, on Jan. 21, 2009, 1 page. cited by applicant
Tribble D E., “An Improved Sump Drain-Irrigation Device of Simple Construction,” Archives of Surgery, vol. 105, Sep. 1972, pp. 511-513. cited by applicant
Usupov Y. N., et al., “Active Wound Drainage,” Russian Journal: Vestnik Khirurgii, Apr. 1987 (p. 42-45), Perspectives in Wound Care, BlueSky Publishing, pp. 8-10. cited by applicant
Venturi M L., et al., “Mechanisms and Clinical Applications of the Vacuum-Assisted Closure (VAC) Device,” Am J Clin Dermatol, 2005, vol. 6, No. 3, pp. 185-194. cited by applicant
Viljanto J., et al., “Local Hyperalimentation of Open Wounds,” Br. J. Surg., vol. 63, 1976, pp. 427-430. cited by applicant
Webb L X., “New Techniques in Wound Management: Vacuum-Assisted Wound Closure,” Journal of the American Academy of Orthopaedic Surgeons, vol. 10, No. 5, Sep./Oct. 2002, pp. 303-311. cited by applicant
Westaby S., et al., “A Wound Irrigation Device,” The Lancet, Sep. 2, 1978, pp. 503-504. cited by applicant
Withdrawal of an Opposition for the European Patent No. 2699276, dated Jul. 31, 2018, in 3 pages. cited by applicant
Wooding-Scott M., et al., “No Wound is Too Big for Resourceful Nurses,” RN, Dec. 1988, pp. 22-25. cited by applicant
Datasheet for the Decision for Oct. 14, 2021 cited during the Appeal Procedure for European Patent No. 2563308, mailed on Dec. 17, 2021, 24 pages. cited by applicant
Boards of Appeal—Letter of the Opponent dated Jan. 20, 2023 for European Patent No. 2821035 mailed on Jan. 25, 2023, 52 pages. cited by applicant
Jack L., et al., “Training Curriculum for Alternative Clothes Cleaning. vol. 1: Curriculum and vol. II: Instructor's Manual and Presentation Materials,” Massachusetts Toxics Use Reduction Institute, 1997, 306 pages. cited by applicant
Reply of the Patent Proprietor to the Notice of Opposition for European Patent No. 3628289, mailed on Dec. 28, 2022, 34 pages. cited by applicant
Reply to Appeal for European Patent No. 2821035 mailed on Feb. 1, 2023, 7 pages. cited by applicant
Statement of Grounds of Appeal by Appellant, re the Opposition of European Patent No. EP2563308, mailed on Dec. 5, 2017, 26 pages. cited by applicant
Statement of Grounds of Appeal for the European Patent No. 2821035, mailed on Sep. 28, 2022, 73 pages. cited by applicant

Termination of the Opposition Proceedings with Revocation of the Patent for European Patent No. 2563306, dated Sep. 13, 2022, 1 page. cited by applicant
Brief Communication—Oral Proceedings and Letter from the opponent for European Patent No. 2821035, mailed on Mar. 7, 2022, 3 pages. cited by applicant
Brief Communication—Oral proceedings and Letter from the Opponent, re the Opposition of European Patent No. 2821035, mailed on Feb. 9, 2022, 4 pages. cited by applicant
Brief Communication—Oral proceedings and Letter from the Opponent, re the Opposition of European Patent No. 2821035, mailed on Jan. 28, 2022, 31 pages. cited by applicant
Brief Communication—Oral proceedings, Letter from the proprietor of the patent and Main and Auxiliary requests, re the Opposition of European Patent No. 2821035, mailed on Feb. 8, 2022, 30 pages. cited by applicant
Claims as Originally filed in Divisional Application 19201361.3, filed in the Opposition of European Patent No. 3628289 on Jul. 18, 2022, 2 pages. cited by applicant
Communication of a Notice of Opposition for the European patent No. 3628289, dated Aug. 12, 2022, 56 pages. cited by applicant
Communication of a Notice of Opposition including Statement of Facts and Evidence for European Patent No. 3628289, dated Jul. 22, 2022, 21 pages. cited by applicant
Communication of a Notice of Opposition including Statement of Facts and Evidence for European Patent No. 3628289, dated Jul. 22, 2022, 41 pages. cited by applicant
Decision revoking the European Patent (Art. 101(3)(b) EPC) for European Patent No. 2821035, mailed on May 19, 2022, 16 pages. cited by applicant
Information about the result of oral proceedings for European Patent No. 2821035, mailed on Apr. 6, 2022, 2 pages. cited by applicant
KCI Licensing, Inc., “V.A.C. Drape® Instructions For Use,” Oct. 2003, 2 pages. cited by applicant
Merriam-Webster, Definition of “Port”, retrieved from URL: https://www.merriam-webster.com/dictionary/port, on Jun. 20, 2022, 3 pages. cited by applicant
Moore A.T., et al., “The Swelling of Cotton in Water: A Microscopical Study,” Textile Research Journal, Sep. 1950, pp. 620-630. cited by applicant
Provision of the minutes in accordance with Rule 124(4) EPC for European Patent No. 2821035, mailed on May 17, 2022, 9 pages. cited by applicant
Silverlon, “Negative Pressure Contact Dressings,” retrieved from Internet URL: https://www.silverlon.com/products/negative-pressure-contact-dressings, accessed on Jan. 24, 2022, 10 pages. cited by applicant
Smith & Nephew, “Allevyn: Technical Information Sheet,” Dec. 2013, 2 pages. cited by applicant
“System,” Cambridge English Dictionary, Retrieved from the Internet: URL: https://dictionary.cambridge.org/dictionary/english/system on Jul. 28, 2022, 1 page. cited by applicant
Termination of Opposition Proceedings for European Patent No. 2563308, dated Jan. 25, 2022, 1 page. cited by applicant
Wikipedia, “Breathability,” Retrieved from the Internet: URL: https://en.wikipedia.org/wiki/Breathability on Aug. 2 2022, 3 pages. cited by applicant
Withdrawal of an appeal for the European Patent No. 2563306, mailed on Apr. 29, 2022, 3 pages. cited by applicant
BASF., “Thermoplastic Polyurethane Elastomers,” Elastollan—Material Properties, 2017, 52 pages, Retrieved from the Internet: URL: https://www.basf.com/kr/documents/ko/product/Elastollan_Material%20Properties.pdf. cited by applicant
BASF., “Thermoplastic Polyurethane Elastomers (TPU),” Elastollan—Product Range, 2016, 40 pages, Retrieved from the Internet: URL: https://www.basf.com/kr/documents/ko/product/Thermoplastic%20Polyurethane_Elastollan_Product%20Range.pdf cited by applicant
3M United States, “3M™ Tegaderm™ Transparent Film Dressing Frame Style,” retrieved Jul. 21, 2022 from https://www.3m.com/3M/en_US/p/d/b00035590/, 4 pages. cited by applicant
Allevyn Adhesive., “SMTL Dressings Datacard,” Smith and Nephew Healthcare Ltd, last modified May 11, 2007, 3 pages. cited by applicant
Annex to the Communication filed in the opposition against European Patent No. 4023197, mailed on Jul. 26, 2024, 27 pages. cited by applicant
Brief Communication—Letter from the Opponent O1 of Jun. 28, 2024 filed in the Opposition against European Patent No. 4023197, dated Jul. 4, 2024, 25 pages. cited by applicant

Brief Communication—Letter from the Proprietor of the Patent of Apr. 29, 2024 for European Patent No. 4023197, mailed May 3, 2024, 56 pages. cited by applicant
Communication of a Notice of Opposition for the European patent No. 4023197, dated Dec. 20, 2023, 50 pages. cited by applicant
Communication of a Notice of Opposition for the European patent No. 4023197, dated Jan. 8, 2024, 44 pages. cited by applicant
Communication of a Notice of Opposition for the European patent No. 4023197, dated Nov. 20, 2023, 49 pages. cited by applicant
Communication of Further Notices of Opposition Pursuant to Rule 79(2) for the European patent No. 4023197, dated Jan. 12, 2024, 98 pages. cited by applicant
Communication of the Board of Appeal pursuant to Article 15(1) of the Rules of Procedure filed in the Opposition against European Patent No. 2821035, mailed on Jul. 23, 2024, 8 pages. cited by applicant
Definition of "Strand," Collins English Dictionary, accessed Apr. 11, 2024, 15 pages, URL: https://www.collinsdictionary.com/dictionary/english/strand. cited by applicant
Definition of "System," Cambridge Dictionary, accessed Apr. 11, 2024, 8 pages, URL:https://dictionary.cambridge.org/dictionary/english/system. cited by applicant
"Search Results from UK Intellectual Property Office Trade Mark Register," filed in the Opposition for the European Patent No. 4023197 on Dec. 15, 2023, 3 pages. cited by applicant
"Handbook of Medical Textiles," Woodhead Publishing Limited, 2011, 41 pages. cited by applicant
Hutten I.M., "Introduction to Nonwoven Filter Media", in Handbook of Nonwoven Filter Media, 2007, 1 page. cited by applicant
Riedel E., et al., "Verbandstoff-Fibel," 5. Auflage, Stuttgart, 1995, Seiten 23-35 and 82-95. cited by applicant
Summons to Attend Oral Proceedings Pursuant to Rule 115(1) EPC for the Opposition of European Patent No. 3628289, mailed on Feb. 9, 2024, 12 pages. cited by applicant
"Technical datasheet for the Chem-PositeTM 11C—450 material," Emerging Technologies, filed in the Opposition for the European Patent No. 4023197 on Apr. 29, 2024 (publication date unknown), 1 page. cited by applicant
"Strand," Wiktionary, last edited Mar. 25, 2024 (accessed Apr. 11, 2024), 10 pages, URL: https://en.wiktionary.org/wiki/strand. cited by applicant
Transmittal of Decision Summons for the Opposition of European Patent No. 3628289, mailed on Jan. 19, 2024, 12 pages. cited by applicant
Brief Communication—the EPO-form 2310 filed in the Opposition against European Patent No. 4023197, mailed on Aug. 19, 2024, 29 pages. cited by applicant
Brief Communication—Letter from the Opponent O1 of Oct. 9, 2024 filed in the Opposition against European Patent No. 3628289, dated Oct. 16, 2024, 3 pages. cited by applicant
Brief Communication—Letter from the Opponent O1 of Sep. 6, 2024 filed in the Opposition against European Patent No. 3628289, dated Sep. 16, 2024, 4 pages. cited by applicant
Brief communication—Letter from the proprietor of the patent of Sep. 11, 2024 for European Patent No. 3628289, mailed on Sep. 18, 2024, 54 pages. cited by applicant
Brief communication—Opposition Proceedings for European Patent No. 3628289, mailed on Dec. 2, 2024, 35 pages. cited by applicant
Brief Communication of Opposition—Letter from the proprietor from the patent to the opponent Simmons & Simmons for the Opposition of European Patent No. 3628289, mailed on Sep. 13, 2024, 33 pages. cited by applicant
Information concerning the result of oral proceedings for the Opposition of European Patent No. 3628289, dated Nov. 11, 2024, 2 pages. cited by applicant
Notification of Cancellation of Oral Proceedings for Oct. 10, 2024, filed in the Opposition against European Patent No. 2821035, mailed on Aug. 16, 2024, 2 pages. cited by applicant
Withdrawal of Appeal—Letter of the Patent Proprietor dated Aug. 9, 2024 filed in the Opposition against European Patent No. 2821035, mailed on Aug. 14, 2024, 4 pages. cited by applicant
Written Submission in Preparation for the Oral Proceedings, for the Opposition of European Patent No. 3628289, dated Oct. 21, 2024, 9 pages. cited by applicant
Brief Communication—Letter from the Opponent O1 of Mar. 14, 2025 filed in the Opposition against European Patent No. 4023197, dated Mar. 19, 2025, 36 pages. cited by applicant
Brief Communication—Letter from the Opponent O2 of Mar. 19, 2025 filed in the Opposition against European Patent No. 4023197, dated Mar. 24, 2025, 22 pages. cited by applicant

Brief Communication—Letter from the Opponent O3 of Apr. 25, 2025 filed in the Opposition against European Patent No. 4023197, dated May 2, 2025, 08 pages. cited by applicant
Brief Communication—Letter from the Opponent O3 of Mar. 19, 2025 filed in the Opposition against European Patent No. 4023197, dated Mar. 24, 2025, 24 pages. cited by applicant
Brief Communication—Letter from the proprietor of the patent of Mar. 19, 2025 filed in the Opposition against European Patent No. 4023197, mailed on Mar. 24, 2025, 45 pages. cited by applicant
Brief Communication—Letter from the Proprietor of the Patent of May 13, 2025 for European Patent No. 4023197, mailed May 14, 2025, 06 pages. cited by applicant
Information about the Result of Oral Proceedings, for the European Patent No. 4023197, dated May 19, 2025, 2 pages. cited by applicant
Statement of Grounds of Appeal for European Patent No. 3628289, mailed on Apr. 2, 2025, 115 pages. cited by applicant
"The Allevyn Wound Dressings Range" 1 page, Product brochure obtained from: https://www.smith-nephew.com/en-pb/health-care-professionals/products/advanced-wound-management/allevynborder-gilobal-new#relatedtechnologies. cited by applicant
The Wayback Machine, "Rayon Fiber," Extract from Conservation and Art Material Encyclopaedia Online, on Mar. 13, 2006, 1 page. cited by applicant
Affidavit of Dr. Viktoria Skeppstedt filed in the Opposition against European Patent No. 4223265, dated May 19, 2025, 2 pages. cited by applicant
Anonymous, "3M™ Tegaderm™ Transparent Film Dressings—Product Profile," Retrieved from http://multimedia.3m.com/mws/media/447983O/tegaderm-transparent-film-dressing-brochure.pdf, Jan. 1, 2012, 8 pages. cited by applicant
Communication of a Notice of Opposition for European Patent No. 4223265, mailed on May 26, 2025, 67 pages. cited by applicant
Decision revoking the European Patent (Art. 101 (3)(b) EPC) for European Patent No. 4023197, mailed on Jun. 5, 2025, 51 pages. cited by applicant
Notice of Opposition for European Patent No. 4223265, mailed on May 28, 2025, 62 pages. cited by applicant
Phan T T., et al., "Evaluation of cell culture on the polyurethane-based membrane (Tegaderm): implication for tissue engineering of skin," Cell Tissue Bank. 2005; 6(2), 1 page. cited by applicant
Russell S.J., et al., "18—Anisotropic Fluid Transmission in Nonwoven Wound Dressings," Proceedings of the 2nd International Conference, Aug. 24, 25, 1999, Woodhead Publishing Series in Textiles, 2001, 1 page. cited by applicant
The Wayback Machine, "Cellulose fiber," Extract from Wikipedia, the free encyclopedia, Retrieved from the Internet: URL: http://en.wikipedia.org/wiki/Cellulose_fiber, Online, on 2009, 1 page. cited by applicant
Wikipedia, Definition of "Rayon" being Regenerated Cellulose, Retrieved from the Internet: https://en.wikipedia.org/wiki/Rayon, on Feb. 7, 2025, 1 page. cited by applicant
Reply to Appeal for the Opposition of European Patent No. 3628289, mailed on Aug. 5, 2025, 68 pages. cited by applicant
Reply to Appeal for the Opposition of European Patent No. 3628289, mailed on Aug. 7, 2025, 16 pages. cited by applicant
Reply to Appeal for the Opposition of European Patent No. 3628289, mailed on Aug. 8, 2025, 88 pages. cited by applicant

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Primary Examiner: Eisenberg; Rebecca E

Assistant Examiner: Wrubleski; Matthew

Attorney, Agent or Firm: Knobbe, Martens, Olson & Bear, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/372,252, filed on Jul. 9, 2021, which is a continuation of U.S. patent application Ser. No. 16/118,339, filed on Aug. 30, 2018 and issued as U.S. Pat. No. 11,058,587, which is a continuation of

U.S. patent application Ser. No. 15/633,670, filed on Jun. 26, 2017 and issued as U.S. Pat. No. 10,159,604, which is a continuation of U.S. patent application Ser. No. 14/715,399, filed on May 18, 2015 and issued as U.S. Pat. No. 9,808,561, which is a continuation of U.S. patent application Ser. No. 13/092,042, filed on Apr. 21, 2011 and issued as U.S. Pat. No. 9,061,095, which claims priority to Great Britain Patent Application No. 1006986.2, filed Apr. 27, 2010; Great Britain Patent Application No. 1006983.9, filed Apr. 27, 2010; Great Britain Patent Application No. 1006985.4, filed Apr. 27, 2010; Great Britain Patent Application No. 1006988.8, filed Apr. 27, 2010; and Great Britain Patent Application No. 1008347.5, filed May 19, 2010; all of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) Embodiments of the present invention relate to methods and apparatuses for dressing and treating a wound with topical negative pressure (TNP) therapy. In particular, but not exclusively, embodiments disclosed herein relate to a wound dressing for providing protection to a wound site, in which the wound dressing acts as a buffer to help prevent compression or shear forces exerted on the wound dressing, for example due to patient movement, from harming a healing wound. Embodiments of the wound dressing may act as a waste canister to collect and store wound exudate removed from a wound site, and also relate to the management of solid build-up in a wound dressing covering a wound site whilst TNP therapy is applied. Further, embodiments disclosed herein relate to a method and suction port for applying negative pressure to a wound dressing and a method of manufacturing a suction port and wound dressing.

Description of the Related Art

(2) Many different types of wound dressing are known for aiding in the healing process of a human or animal. These different types of wound dressing include many different types of materials and layers, for example, gauze, pads, foam pads or multi-layer wound dressings.

(3) In addition, TNP therapy, sometimes referred as vacuum assisted closure or negative pressure wound therapy, has recently been proposed as a successful mechanism for improving the healing rate of a wound. Such therapy is applicable to a broad range of wounds such as incisional wounds, open wounds and abdominal wounds or the like.

(4) TNP therapy assists in the closure and healing of wounds by reducing tissue oedema; encouraging blood flow; stimulating the formation of granulation tissue; removing excess exudates and may reduce bacterial load and thus, infection to the wound. Furthermore, TNP therapy permits less outside disturbance of the wound and promotes more rapid healing.

(5) During TNP therapy, a suction source such as a vacuum pump or the like is utilized to create a negative pressure region. That is to say, a region where an experienced pressure is below that of the surroundings. Wound exudate and other potentially harmful material is extracted from the wound region and must be stored for later disposal. A problem associated with many known techniques is that a separate canister must be provided for storage of such exudate. Provision of such canisters is costly and bulky and prone to failure.

(6) A proposal has been suggested to store extracted wound exudate in the wound dressing itself that is used to cover a wound site and create the wound chamber region where negative pressure is established. However, it is known that many different wound types can exude high flow rates of exudate and therefore storage of exuding material in a wound dressing can be problematical since the wound dressing will only have a limited capacity for fluid uptake before a dressing change is required. This can limit a time of use between dressing changes and can prove costly if many wound dressings are required to treat a given wound.

(7) It has been suggested as a solution to this problem, that a liquid impermeable moisture vapor permeable cover layer can be utilized as an uppermost cover layer for the wound dressing. The air impermeable nature of the cover layer provides a sealing layer over the wound site so that negative pressure can be established below the dressing in the region of the wound. The moisture vapor permeability of this covering layer is selected so that liquid can constantly evaporate away from the top of the dressing. This means that as therapy is continued the dressing does not have to take up and hold all liquid exuding from the wound. Rather, some liquid is constantly escaping in the form of moisture vapor from the upper environs of the dressing.

(8) Whilst such dressings work well in practice, the continuous evaporation of moisture vapor from the dressing can lead to the problem of crust formation in the dressing. That is to say, because of the continuous drawing of liquid away from the wound site solid particulate matter is more prone to formation and accumulation in the dressing. Under certain circumstances the build-up of such solid material can lead to blockages forming in the wound dressing in the flowpath between the wound and the source of negative

pressure. This can potentially cause problems in that therapy may need to be halted to change a dressing if the blockages reach a critical level.
(9) Further, there is much prior art available relating to the provision of apparatuses and methods of use thereof for the application of TNP therapy to wounds together with other therapeutic processes intended to enhance the effects of the TNP therapy.
(10) It will be appreciated that from time to time accidents may happen to patients undergoing negative pressure wound therapy. Such accidents might cause short term or long term forces to be applied to a dressing covering a wound. Alternatively patient movement may bring the patient and any dressing covering a healing wound into contact with an external object. In such occurrences compressive forces or lateral forces may occur. Such force can cause disturbance of a wound bed which can damage a wound site. A particular cause for concern is during the treatment of skin graft wounds. Under such conditions lateral forces can entirely upset or tear apart a healing skin graft region.
SUMMARY OF THE INVENTION
(11) It is an aim of certain embodiments of the present invention to at least partly mitigate the above-mentioned problems.
(12) It is an aim of certain embodiments of the present invention to provide a method for providing negative pressure at a wound site to aid in wound closure and healing in which wound exudate drawn from a wound site during the therapy is collected and stored in a wound dressing.
(13) It is an aim of certain embodiments of the present invention to provide a wound dressing having an increased capacity for absorbing wound exudate reducing the frequency with which the dressings must be changed.
(14) It is further an aim of certain embodiments of the invention to manage the movement of wound exudate through a dressing to avoid blockages occurring that lead to reduced life of the dressing.
(15) It is an aim of certain embodiments of the present invention to provide a wound dressing having an increased capacity to absorb compressive forces exerted on the wound dressing.
(16) It is an aim of certain embodiments of the present invention to provide a wound dressing having an increased capacity to prevent shear forces from an outer surface of a wound dressing from being translated into corresponding shear forces at a wound site.
(17) It is an aim of certain embodiments of the present invention to provide a wound dressing which can “give” in a direction perpendicular to and parallel to a wound site surface even when the dressing experiences negative pressure.
(18) It is an aim of certain embodiments of the present invention to provide a wound dressing able to be used with topical negative pressure therapy which helps maintain an open flow path so that therapy can be continued unhindered by blockages caused by build-up of solid matter.
(19) It is an aim of certain embodiments of the present invention to provide a method and apparatus for treating a wound with topical negative pressure therapy by preventing blockage of a flowpath region of a wound dressing.
(20) Embodiments disclosed herein are directed toward the treatment of wounds with TNP. In particular, certain embodiments disclose a wound dressing capable of absorbing and storing wound exudate in conjunction with a pump, for example a miniaturized pump. Some wound dressing embodiments further comprise a transmission layer configured to transmit wound exudates to an absorbent layer disposed in the wound dressing. Additionally, some embodiments provide for a port or other fluidic connector configured to retain wound exudate within the wound dressing while transmitting negative pressure to the wound dressing.
(21) According to a first embodiment of the present invention there is provided a wound treatment apparatus comprising: a wound dressing comprising: a transmission layer comprising a 3D knitted or fabric material configured to remain open upon application of negative pressure to the wound dressing; an absorbent layer for absorbing wound exudate, the absorbent layer overlying the transmission layer; a cover layer overlying the absorbent layer and comprising an orifice, wherein the cover layer is moisture vapor permeable; a pump; and a suction port for applying negative pressure to the wound dressing for the application of topical negative pressure at a wound site, the suction port comprising: a connector portion for connecting the suction port to the pump; a sealing surface for sealing the suction port to the cover layer of the wound dressing; and a liquid impermeable gas permeable filter element arranged to prevent a liquid from entering the connector portion.
(22) According to a second embodiment of the present invention there is provided a method for the treatment of a wound comprising: providing a wound dressing comprising: a transmission layer comprising a 3D knitted or fabric material; an absorbent layer for absorbing wound exudate, the absorbent layer overlying the

transmission layer; a cover layer overlying the absorbent layer and comprising an orifice, wherein the cover layer is moisture vapor permeable; positioning the dressing over a wound site to form a sealed cavity over the wound site; and applying negative pressure to the wound site to draw fluid through the transmission layer into the absorbent layer.

(23) According to a another embodiment of the present invention there is provided a wound dressing for providing protection at a wound site, comprising: a transmission layer comprising a first surface and a further surface spaced apart from the first surface by a relax distance in a relaxed mode of operation; and a plurality of spacer elements extending between the first and further surfaces and, in a forced mode of operation, locatable whereby the first and further surfaces are spaced apart by a compression distance less than the relax distance.

(24) According to a one embodiment of the present invention there is provided a method for providing protection at a wound site, comprising: locating a wound dressing comprising a transmission layer over a wound site; and responsive to a force on the wound dressing, displacing a plurality of spacer elements extending between a first surface and a further surface of the transmission layer whereby; a distance between the first and further surfaces is reduced as the spacer elements are displaced.

(25) According to another embodiment of the present invention there is provided an apparatus for dressing a wound for the application of topical negative pressure at a wound site, comprising: a liquid and gas permeable transmission layer; an absorbent layer for absorbing wound exudate, the absorbent layer overlying the transmission layer; a gas impermeable cover layer overlying the absorbent layer and comprising a first orifice, wherein the cover layer is moisture vapor permeable.

(26) According to a further embodiment of the present invention there is provided a method of applying TNP at a wound site, comprising the steps of: applying negative pressure at an orifice of a cover layer of a wound dressing, a peripheral region around the wound site being sealed with the wound dressing, such that air and wound exudate are drawn towards the orifice; collecting wound exudate, drawn from the wound site, through a transmission layer of the wound dressing, in an absorbent layer of the wound dressing; and transpiring a water component of the wound exudate collected in the absorbent layer through the cover layer of the wound dressing.

(27) According to an additional embodiment of the present invention there is provided apparatus for dressing a wound for the application of topical negative pressure at a wound site, comprising: a liquid and gas permeable transmission layer; an absorbent layer for absorbing wound exudate; a gas impermeable cover layer overlying the absorbent layer and the transmission layer, the cover layer comprising an orifice connected to the transmission layer; and at least one element configured to reduce the rate at which wound exudate moves towards the orifice when a negative pressure is applied at the orifice.

(28) According to another embodiment of the present invention there is provided a method of applying TNP at a wound site, comprising the steps of: applying negative pressure at an orifice of a cover layer of a wound dressing, a peripheral region around the wound site being sealed with the wound dressing such that air and wound exudate move towards the orifice; collecting wound exudate, from the wound site, through a transmission layer of the wound dressing, in an absorbent layer of the wound dressing; and reducing the rate at which wound exudate moves towards the orifice.

(29) According to still another embodiment of the present invention there is provided apparatus for dressing a wound for the application of topical negative pressure at a wound site, comprising: an absorbent layer for absorbing wound exudate; a gas impermeable cover layer overlying the absorbent layer the cover layer comprising at least one orifice configured to allow negative pressure to be communicated through the cover layer in at least two spaced apart regions.

(30) According to an additional embodiment of the present invention there is provided a method of applying TNP at a wound site, comprising the steps of: sealing a cover layer of a wound dressing around the wound site; applying negative pressure at at least one orifice in the cover layer, said at least one orifice configured to allow negative pressure to be communicated through the cover layer in at least two spaced apart regions; and collecting wound exudate, from the wound site, in an absorbent layer of the wound dressing.

(31) According to one embodiment of the present invention there is provided a suction port for applying negative pressure to a wound dressing for the application of topical negative pressure at a wound site, the suction port comprising: a connector portion for connecting the suction port to a source of negative pressure; a sealing surface for sealing the suction port to a cover layer of a wound dressing; and a liquid impermeable gas permeable filter element arranged to prevent a liquid entering the connector portion.

(32) According to an additional embodiment of the present invention there is provided a method of

communicating negative pressure to a wound dressing for the application of topical negative pressure at a wound site, comprising the steps of: applying negative pressure at a connecting portion of a suction port sealed around a perimeter of an orifice in a cover layer of the wound dressing; filtering gas drawn from within the wound dressing through a liquid impermeable gas permeable filter element of the suction port.
(33) According to another embodiment of the invention there is provided a method of manufacturing a suction port for applying negative pressure to a wound dressing for the application of topical negative pressure at a wound site, the suction port having a connector portion for connecting the suction port to a source of negative pressure and a sealing surface for sealing the suction port to a cover layer of a wound dressing, the method comprising: disposing a liquid impermeable gas permeable filter element of the suction port at a location to prevent a liquid entering the connector portion.
(34) According to yet another embodiment of the present invention there is provided apparatus for the application of TNP therapy to a wound site, comprising: a first layer comprising a plurality of openings each having a first open area; a further layer spaced apart from the first layer comprising a plurality of further openings each having a further open area; and an air impermeable, moisture vapor permeable cover layer over the first and further layers; wherein a region between the first and further layers comprises a portion of a flow path for air and/or wound exudate flowing from a wound site and said first open area is less than said further open area.
(35) According to still another embodiment of the present invention there is provided a method of applying TNP therapy to a wound site, comprising: via a vacuum pump in fluid communication with a wound dressing located over a wound site, applying a negative pressure at the wound site; and as liquid evaporates through a cover layer of the dressing, preventing blockage of a fluid flowpath region of the wound dressing.
(36) Certain embodiments provide a wound dressing which even when under negative pressure conditions is able to provide further “give” to buffer compression forces from harming a wound.
(37) Certain embodiments provide a wound dressing able to disconnect shear forces applied to the dressing from the wound site covered by the dressing. As a result damage to the wound can be wholly or at least partially avoided.
(38) Certain embodiments provide the advantage that a wound site can be covered with a wound dressing which is simultaneously able to deliver negative pressure wound therapy to a wound site, collect exudate and provide protection from forces operating on the dressing.
(39) Certain embodiments provide the advantage that forces operating on a dressing can be offset by dissipating loads operating over a relatively small distance on an upper layer of the dressing to a relatively larger area on a lower surface of the dressing. The force is thus dissipated over a larger area thus reducing the effect of the force.
(40) Certain embodiments provide the advantage that a wound dressing can be used to collect wound exudate generated during a negative pressure therapy process, whilst extending the useful lifetime of the dressing by transpiring a water component of the wound exudate. A pump remote from the wound dressing can be connected to the wound dressing and reused whilst the wound dressing itself is used to collect wound exudate and may then be disposed of after use.
(41) Certain embodiments provide a wound dressing and/or method of applying topical negative pressure in which a flowpath through a wound dressing is kept open so that therapy can be continued for as long as desired by a care giver.
(42) Certain embodiments prevent solid material, which may cause a blockage, from entering a flowpath region in the wound dressing by using a layer of the dressing to act as a bar to such material.
(43) Certain embodiments prevent build-up of solid material in a flowpath region of a wound dressing by ensuring that any solid material that enters into that flowpath region can always escape into a further region of the dressing.
(44) Certain embodiments provide the advantage that the build-up of solid material in a flowpath in a wound dressing is avoided by having an absorbent layer close to the flowpath region store liquid over time. This helps keep the environment of the flowpath region moist which helps avoid crusting.
(45) Certain embodiments provide the advantage that a wound dressing can be used to collect wound exudate generated during a negative pressure therapy process, whilst extending the useful lifetime of the dressing by transpiring a water component of the wound exudate. A pump remote from the wound dressing can be connected to the wound dressing and reused whilst the wound dressing itself is used to collect wound exudate and may then be disposed of after use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of the present invention will now be described hereinafter, by way of example only, with reference to the accompanying drawings in which:
(2) FIG. **1**A illustrates a wound dressing;
(3) FIG. **1**B illustrates another embodiment of a wound dressing;
(4) FIG. **2** illustrates a top view of a wound dressing;
(5) FIG. **3** illustrates a top view of a wound dressing including baffle elements;
(6) FIG. **4** illustrates a top view of a further wound dressing including baffle elements;
(7) FIG. **5** illustrates a baffle element according to one embodiment;
(8) FIG. **6** illustrates a top view of a wound dressing including a single baffle element;
(9) FIG. **7** illustrates a top view of a wound dressing including an air channel;
(10) FIG. **8** illustrates a top view of a wound dressing including two air channels;
(11) FIG. **9** illustrates a top view of a wound dressing including two orifices in a cover layer coupled through a fluid communication passage;
(12) FIG. **10** illustrates an embodiment of the fluid communication passage;
(13) FIG. **11** illustrates a top view of a suction port;
(14) FIG. **12** illustrates a suction port including a filter element;
(15) FIG. **13** illustrates a further suction port including a filter element;
(16) FIGS. **14**A-L illustrate a range of exemplifying configurations of baffle elements in a wound dressing;
(17) FIG. **15** illustrates an exemplifying configuration of vias in a transmission layer of a wound dressing;
(18) FIG. **16** illustrates a top view of a wound dressing including an elongate orifice in a cover layer;
(19) FIG. **17** illustrates a transmission layer in a relaxed mode of operation;
(20) FIG. **18** illustrates a transmission layer in a forced mode of operation;
(21) FIG. **19** illustrates pressure offsetting;
(22) FIG. **20** illustrates a transmission layer and overlying absorbent layer in a relaxed mode of operation;
(23) FIG. **21** illustrates an absorbent layer and transmission layer experiencing a compressive force;
(24) FIG. **22** illustrates an absorbent layer and transmission layer experiencing a shear force;
(25) FIG. **23** illustrates a cross-section of a region of an embodiment of a wound dressing;
(26) FIG. **24** illustrates a lower layer of a transmission layer used in an embodiment of a wound dressing;
(27) FIG. **25** illustrates an upper layer of a transmission layer used in an embodiment of a wound dressing;
(28) FIG. **26** illustrates a lower surface of a transmission layer used in another embodiment of a wound dressing;
(29) FIG. **27** illustrates an upper surface of a transmission layer used in another embodiment of a wound dressing;
(30) FIG. **28** illustrates an embodiment of a wound treatment system; and,
(31) FIGS. **29**A-D illustrate the use and application of an embodiment of a wound treatment system onto a patient.
(32) In the drawings like reference numerals refer to like parts.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(33) FIGS. **1**A-B illustrate cross sections through a wound dressing **100** according to an embodiment of the invention. A plan view from above the wound dressing **100** is illustrated in FIG. **2** with the line A-A indicating the location of the cross section shown in FIGS. **1**A and **1**B. It will be understood that FIGS. **1**A-B illustrate a generalized schematic view of an apparatus **100**. It will be understood that embodiments of the present invention are generally applicable to use in TNP therapy systems. Briefly, negative pressure wound therapy assists in the closure and healing of many forms of “hard to heal” wounds by reducing tissue oedema; encouraging blood flow and granular tissue formation; removing excess exudate and may reduce bacterial load (and thus infection risk). In addition, the therapy allows for less disturbance of a wound leading to more rapid healing. TNP therapy systems may also assist on the healing of surgically closed wounds by removing fluid and by helping to stabilize the tissue in the apposed position of closure. A further beneficial use of TNP therapy can be found in grafts and flaps where removal of excess fluid is important and close proximity of the graft to tissue is required in order to ensure tissue viability.
(34) The wound dressing **100** can be located over a wound site to be treated. The dressing **100** forms a sealed

cavity over the wound site. It will be appreciated that throughout this specification reference is made to a wound. In this sense it is to be understood that the term wound is to be broadly construed and encompasses open and closed wounds in which skin is torn, cut or punctured or where trauma causes a contusion. A wound is thus broadly defined as any damaged region of tissue where fluid may or may not be produced. Examples of such wounds include, but are not limited to, incisions, lacerations, abrasions, contusions, burns, diabetic ulcers, pressure ulcers, stoma, surgical wounds, trauma and venous ulcers or the like.

(35) In some embodiments, it may be preferable for the wound site to be filled partially or completely with a wound packing material. This wound packing material is optional, but may be desirable in certain wounds, for example deeper wounds. The wound packing material can be used in addition to the wound dressing **100**. The wound packing material generally may comprise a porous and conformable material, for example foam (including reticulated foams), and gauze. Preferably, the wound packing material is sized or shaped to fit within the wound site so as to fill any empty spaces. The wound dressing **100** may then be placed over the wound site and wound packing material overlying the wound site. When a wound packing material is used, once the wound dressing **100** is sealed over the wound site, TNP is transmitted from a pump through the wound dressing **100**, through the wound packing material, and to the wound site. This negative pressure draws wound exudate and other fluids or secretions away from the wound site.

(36) It is envisaged that the negative pressure range for the apparatus embodying the present invention may be between about −20 mmHg and −200 mmHg (note that these pressures are relative to normal ambient atmospheric pressure thus, −200 mmHg would be about 560 mmHg in practical terms). In one embodiment, the pressure range may be between about −40 mmHg and −150 mmHg. Alternatively a pressure range of up to −75 mmHg, up to −80 mmHg or over −80 mmHg can be used. Also in other embodiments a pressure range of below −75 mmHg could be used. Alternatively a pressure range of over −100 mmHg could be used or over −150 mmHg.

(37) It will be appreciated that according to certain embodiments of the present invention the pressure provided may be modulated over a period of time according to one or more desired and predefined pressure profiles. For example such a profile may include modulating the negative pressure between two predetermined negative pressures P**1** and P**2** such that pressure is held substantially constant at P**1** for a pre-determined time period T**1** and then adjusted by suitable means such as varying pump work or restricting fluid flow or the like, to a new predetermined pressure P**2** where the pressure may be held substantially constant for a further predetermined time period T**2**. Two, three or four or more predetermined pressure values and respective time periods may be optionally utilized. Other embodiments may employ more complex amplitude/frequency wave forms of pressure flow profiles may also be provided e.g. sinusoidal, sore tooth, systolic-diastolic or the like.

(38) As illustrated in FIGS. **1**A-B a lower surface **101** of the wound dressing **100** is provided by an optional wound contact layer **102**. The wound contact layer **102** can be a polyurethane layer or polyethylene layer or other flexible layer which is perforated, for example via a hot pin process, laser ablation process, ultrasound process or in some other way or otherwise made permeable to liquid and gas. The wound contact layer has a lower surface **101** and an upper surface **103**. The perforations **104** are through holes in the wound contact layer which enables fluid to flow through the layer. The wound contact layer helps prevent tissue ingrowth into the other material of the wound dressing. The perforations are small enough to meet this requirement but still allow fluid through. For example, perforations formed as slits or holes having a size ranging from 0.025 mm to 1.2 mm are considered small enough to help prevent tissue ingrowth into the wound dressing while allowing wound exudate to flow into the dressing. The wound contact layer helps hold the whole wound dressing together and helps to create an air tight seal around the absorbent pad in order to maintain negative pressure at the wound. The wound contact layer also acts as a carrier for an optional lower and upper adhesive layer (not shown). For example, a lower pressure sensitive adhesive may be provided on the underside surface **101** of the wound dressing whilst an upper pressure sensitive adhesive layer may be provided on the upper surface **103** of the wound contact layer. The pressure sensitive adhesive, which may be a silicone, hot melt, hydrocolloid or acrylic based adhesive or other such adhesives, may be formed on both sides or optionally on a selected one or none of the sides of the wound contact layer. When a lower pressure sensitive adhesive layer is utilized this helps adhere the wound dressing to the skin around a wound site.

(39) A layer **105** of porous material is located above the wound contact layer. This porous layer, or transmission layer, **105** allows transmission of fluid including liquid and gas away from a wound site into upper layers of the wound dressing. In particular, the transmission layer **105** ensures that an open air channel can be maintained to communicate negative pressure over the wound area even when the absorbent layer has

absorbed substantial amounts of exudates. The layer should remain open under the typical pressures that will be applied during negative pressure wound therapy as described above, so that the whole wound site sees an equalized negative pressure. The layer **105** is formed of a material having a three dimensional structure. For example, a knitted or woven spacer fabric (for example B altex 7970 weft knitted polyester) or a non-woven fabric could be used. Other materials could of course be utilized, and examples of such materials are described below with respect to FIGS. **23**-**27**.
(40) In some embodiments, the transmission layer comprises a 3D polyester spacer fabric layer including a top layer (that is to say, a layer distal from the wound-bed in use) which is a 84/144 textured polyester, and a bottom layer (that is to say, a layer which lies proximate to the wound bed in use) which is a 100 denier flat polyester and a third layer formed sandwiched between these two layers which is a region defined by a knitted polyester viscose, cellulose or the like monofilament fiber. Other materials and other linear mass densities of fiber could of course be used.
(41) Whilst reference is made throughout this disclosure to a monofilament fiber it will be appreciated that a multistrand alternative could of course be utilized.
(42) The top spacer fabric thus has more filaments in a yarn used to form it than the number of filaments making up the yarn used to form the bottom spacer fabric layer.
(43) This differential between filament counts in the spaced apart layers helps control moisture flow across the transmission layer. Particularly, by having a filament count greater in the top layer, that is to say, the top layer is made from a yarn having more filaments than the yarn used in the bottom layer, liquid tends to be wicked along the top layer more than the bottom layer. In use, this differential tends to draw liquid away from the wound bed and into a central region of the dressing where the absorbent layer helps lock the liquid away or itself wicks the liquid onwards towards the cover layer where it can be transpired.
(44) Preferably, to improve the liquid flow across the transmission layer (that is to say perpendicular to the channel region formed between the top and bottom spacer layers, the 3D fabric is treated with a dry cleaning agent (such as, but not limited to, Perchloro Ethylene) to help remove any manufacturing products such as mineral oils, fats and/or waxes used previously which might interfere with the hydrophilic capabilities of the transmission layer. In some embodiments, an additional manufacturing step can subsequently be carried in which the 3D spacer fabric is washed in a hydrophilic agent (such as, but not limited to, Feran Ice 30 g/l available from the Rudolph Group). This process step helps ensure that the surface tension on the materials is so low that liquid such as water can enter the fabric as soon as it contacts the 3D knit fabric. This also aids in controlling the flow of the liquid insult component of any exudates.
(45) A layer **110** of absorbent material is provided above the transmission layer **105**. The absorbent material which may be a foam or non-woven natural or synthetic material and which may optionally include or be super-absorbent material forms a reservoir for fluid, particularly liquid, removed from the wound site and draws those fluids towards a cover layer **140**. The material of the absorbent layer also prevents liquid collected in the wound dressing from flowing in a sloshing manner. The absorbent layer **110** also helps distribute fluid throughout the layer via a wicking action so that fluid is drawn from the wound site and stored throughout the absorbent layer. This helps prevent agglomeration in areas of the absorbent layer. The capacity of the absorbent material must be sufficient to manage the exudates flow rate of a wound when negative pressure is applied. Since in use the absorbent layer experiences negative pressures the material of the absorbent layer is chosen to absorb liquid under such circumstances. A number of materials exist that are able to absorb liquid when under negative pressure, for example superabsorber material. The absorbent layer **110** may typically be manufactured from ALLEVYN™ foam, Freudenberg 114-224-4 and/or Chem-Posite™11C-450.
(46) In some embodiments, the absorbent layer is a layer of non-woven cellulose fibers having super-absorbent material in the form of dry particles dispersed throughout. Use of the cellulose fibers introduces fast wicking elements which help quickly and evenly distribute liquid taken up by the dressing. The juxtaposition of multiple strand-like fibers leads to strong capillary action in the fibrous pad which helps distribute liquid. In this way, the super-absorbent material is efficiently supplied with liquid. Also, all regions of the absorbent layer are provided with liquid.
(47) The wicking action also assists in bringing liquid into contact with the upper cover layer to aid increase transpiration rates of the dressing.
(48) The wicking action also assists in delivering liquid downwards towards the wound bed when exudation slows or halts. This delivery process helps maintain the transmission layer and lower wound bed region in a moist state which helps prevent crusting within the dressing (which could lead to blockage) and helps

maintain an environment optimized for wound healing.
(49) In some embodiments, the absorbent layer may be an air-laid material. Heat fusible fibers may optionally be used to assist in holding the structure of the pad together. It will be appreciated that rather than using super-absorbing particles or in addition to such use, super-absorbing fibers may be utilized according to certain embodiments of the present invention. An example of a suitable material is the Product Chem-Posite™ 11 C available from Emerging Technologies Inc (ETi) in the USA.
(50) Optionally, according to certain embodiments of the present invention, the absorbent layer may include synthetic stable fibers and/or bi-component stable fibers and/or natural stable fibers and/or super-absorbent fibers. Fibers in the absorbent layer may be secured together by latex bonding or thermal bonding or hydrogen bonding or a combination of any bonding technique or other securing mechanism. In some embodiments, the absorbent layer is formed by fibers which operate to lock super-absorbent particles within the absorbent layer. This helps ensure that super-absorbent particles do not move external to the absorbent layer and towards an underlying wound bed. This is particularly helpful because when negative pressure is applied there is a tendency for the absorbent pad to collapse downwards and this action would push super-absorbent particle matter into a direction towards the wound bed if they were not locked away by the fibrous structure of the absorbent layer.
(51) The absorbent layer may comprise a layer of multiple fibers. Preferably, the fibers are strand-like and made from cellulose, polyester, viscose or the like. Preferably, dry absorbent particles are distributed throughout the absorbent layer ready for use. In some embodiments, the absorbent layer comprises a pad of cellulose fibers and a plurality of super absorbent particles. In additional embodiments, the absorbent layer is a non-woven layer of randomly orientated cellulose fibers.
(52) Super-absorber particles/fibers may be, for example, sodium polyacrylate or carbomethoxycellulose materials or the like or any material capable of absorbing many times its own weight in liquid. In some embodiments, the material can absorb more than five times its own weight of 0.9% W/W saline, etc. In some embodiments, the material can absorb more than 15 times its own weight of 0.9% W/W saline, etc. In some embodiments, the material is capable of absorbing more than 20 times its own weight of 0.9% W/W saline, etc. Preferably, the material is capable of absorbing more than 30 times its own weight of 0.9% W/W saline, etc.
(53) Preferably, the particles of superabsorber are very hydrophilic and grab the fluid as it enters the dressing, swelling up on contact. An equilibrium is set up within the dressing core whereby moisture passes from the superabsorber into the dryer surrounding area and as it hits the top film the film switches and the fluid vapor starts to be transpired. A moisture gradient is established within the dressing to continually remove fluid from the wound bed and ensure the dressing does not become heavy with exudate.
(54) Preferably the absorbent layer includes at least one through hole located so as to underly the suction port. As illustrated in FIGS. **1**A-B a single through hole can be used to produce an opening underlying the port **150**. It will be appreciated that multiple openings could alternatively be utilized. Additionally should more than one port be utilized according to certain embodiments of the present invention one or multiple openings may be made in the super-absorbent layer in registration with each respective port. Although not essential to certain embodiments of the present invention the use of through holes in the super-absorbent layer provide a fluid flow pathway which is particularly unhindered and this is useful in certain circumstances.
(55) Where an opening is provided in the absorbent layer the thickness of the layer itself will act as a stand-off separating any overlying layer from the upper surface (that is to say the surface facing away from a wound in use) of the transmission layer **105**. An advantage of this is that the filter of the port is thus decoupled from the material of the transmission layer. This helps reduce the likelihood that the filter will be wetted out and thus will occlude and block further operation.
(56) Use of one or more through holes in the absorption layer also has the advantage that during use if the absorbent layer contains a gel forming material, such as superabsorber, that material as it expands to absorb liquid, does not form a barrier through which further liquid movement and fluid movement in general cannot pass. In this way each opening in the absorbent layer provides a fluid pathway between the transmission layer directly to the wound facing surface of the filter and then onwards into the interior of the port.
(57) A gas impermeable, but moisture vapor permeable, cover layer **140** extends across the width of the wound dressing. The cover layer, which may for example be a polyurethane film (for example, Elastollan SP9109) having a pressure sensitive adhesive on one side, is impermeable to gas and this layer thus operates to cover the wound and to seal a wound cavity over which the wound dressing is placed. In this way an

effective chamber is made between the cover layer and a wound site where a negative pressure can be established. The cover layer **140** is sealed to the wound contact layer **102** in a border region **200** around the circumference of the dressing, ensuring that no air is drawn in through the border area, for example via adhesive or welding techniques. The cover layer **140** protects the wound from external bacterial contamination (bacterial barrier) and allows liquid from wound exudates to be transferred through the layer and evaporated from the film outer surface. The cover layer **140** typically comprises two layers; a polyurethane film and an adhesive pattern spread onto the film. The polyurethane film is moisture vapor permeable and may be manufactured from a material that has an increased water transmission rate when wet.
(58) The absorbent layer **110** may be of a greater area than the transmission layer **105**, such that the absorbent layer overlaps the edges of the transmission layer **105**, thereby ensuring that the transmission layer does not contact the cover layer **140**. This provides an outer channel **115** of the absorbent layer **110** that is in direct contact with the wound contact layer **102**, which aids more rapid absorption of exudates to the absorbent layer. Furthermore, this outer channel **115** ensures that no liquid is able to pool around the circumference of the wound cavity, which may otherwise seep through the seal around the perimeter of the dressing leading to the formation of leaks.
(59) In order to ensure that the air channel remains open when a vacuum is applied to the wound cavity, the transmission layer **105** must be sufficiently strong and non-compliant to resist the force due to the pressure differential. However, if this layer comes into contact with the relatively delicate cover layer **140**, it can cause the formation of pin-hole openings in the cover layer **140** which allow air to leak into the wound cavity. This may be a particular problem when a switchable type polyurethane film is used that becomes weaker when wet. The absorbent layer **110** is generally formed of a relatively soft, non-abrasive material compared to the material of the transmission layer **105** and therefore does not cause the formation of pin-hole openings in the cover layer. Thus by providing an absorbent layer **110** that is of greater area than the transmission layer **105** and that overlaps the edges of the transmission layer **105**, contact between the transmission layer and the cover layer is prevented, avoiding the formation of pin-hole openings in the cover layer **140**.
(60) The absorbent layer **110** is positioned in fluid contact with the cover layer **140**. As the absorbent layer absorbs wound exudate, the exudate is drawn towards the cover layer **140**, bringing the water component of the exudate into contact with the moisture vapor permeable cover layer. This water component is drawn into the cover layer itself and then evaporates from the top surface of the dressing. In this way, the water content of the wound exudate can be transpired from the dressing, reducing the volume of the remaining wound exudate that is to be absorbed by the absorbent layer **110**, and increasing the time before the dressing becomes full and must be changed. This process of transpiration occurs even when negative pressure has been applied to the wound cavity, and it has been found that the pressure difference across the cover layer when a negative pressure is applied to the wound cavity has negligible impact on the moisture vapor transmission rate across the cover layer.
(61) An orifice **145** is provided in the cover film **140** to allow a negative pressure to be applied to the dressing **100**. A suction port **150** is sealed to the top of the cover film **140** over the orifice **145**, and communicates negative pressure through the orifice **145**. A length of tubing **220** may be coupled at a first end to the suction port **150** and at a second end to a pump unit (not shown) to allow fluids to be pumped out of the dressing. The port may be adhered and sealed to the cover film **140** using an adhesive such as an acrylic, cyanoacrylate, epoxy, UV curable or hot melt adhesive. The port **150** is formed from a soft polymer, for example a polyethylene, a polyvinyl chloride, a silicone or polyurethane having a hardness of 30 to 90 on the Shore A scale.
(62) An aperture or through-hole **146** is provided in the absorbent layer **110** beneath the orifice **145** such that the orifice is connected directly to the transmission layer **105**. This allows the negative pressure applied to the port **150** to be communicated to the transmission layer **105** without passing through the absorbent layer **110**. This ensures that the negative pressure applied to the wound site is not inhibited by the absorbent layer as it absorbs wound exudates. In other embodiments, no aperture may be provided in the absorbent layer **110**, or alternatively a plurality of apertures underlying the orifice **145** may be provided.
(63) As shown in FIG. **1**A, one embodiment of the wound dressing **100** comprises an aperture **146** in the absorbent layer **100** situated underneath the port **150**. In use, for example when negative pressure is applied to the dressing **100**, a wound facing portion of the port **150** may thus come into contact with the transmission layer **105**, which can thus aid in transmitting negative pressure to the wound site even when the absorbent layer **110** is filled with wound fluids. Some embodiments may have the cover layer **140** be at least partly adhered to the transmission layer **105**. In some embodiments, the aperture **146** is at least 1-2 mm larger than

the diameter of the wound facing portion of the port **150**, or the orifice **145**.
(64) A filter element **130** that is impermeable to liquids, but permeable to gases is provided to act as a liquid barrier, and to ensure that no liquids are able to escape from the wound dressing. The filter element may also function as a bacterial barrier. Typically the pore size is 0.2 μm. Suitable materials for the filter material of the filter element **130** include 0.2 micron Gore™ expanded PTFE from the MMT range, PALL Versapore™ 200R, and Donaldson™ TX6628. Larger pore sizes can also be used but these may require a secondary filter layer to ensure full bioburden containment. As wound fluid contains lipids it is preferable, though not essential, to use an oleophobic filter membrane for example 1.0 micron MMT-332 prior to 0.2 micron MMT-323. This prevents the lipids from blocking the hydrophobic filter. The filter element can be attached or sealed to the port and/or the cover film **140** over the orifice **145**. For example, the filter element **130** may be molded into the port **150**, or may be adhered to both the top of the cover layer **140** and bottom of the port **150** using an adhesive such as, but not limited to, a UV cured adhesive.
(65) It will be understood that other types of material could be used for the filter element **130**. More generally a microporous membrane can be used which is a thin, flat sheet of polymeric material, this contains billions of microscopic pores. Depending upon the membrane chosen these pores can range in size from 0.01 to more than 10 micrometers. Microporous membranes are available in both hydrophilic (water filtering) and hydrophobic (water repellent) forms. In some embodiments of the invention, filter element **130** comprises a support layer and an acrylic co-polymer membrane formed on the support layer. Preferably the wound dressing **100** according to certain embodiments of the present invention uses microporous hydrophobic membranes (MHMs). Numerous polymers may be employed to form MHMs. For example, PTFE, polypropylene, PVDF and acrylic copolymer. All of these optional polymers can be treated in order to obtain specific surface characteristics that can be both hydrophobic and oleophobic. As such these will repel liquids with low surface tensions such as multi-vitamin infusions, lipids, surfactants, oils and organic solvents.
(66) MHMs block liquids whilst allowing air to flow through the membranes. They are also highly efficient air filters eliminating potentially infectious aerosols and particles. A single piece of MHM is well known as an option to replace mechanical valves or vents. Incorporation of MHMs can thus reduce product assembly costs improving profits and costs/benefit ratio to a patient.
(67) The filter element **130** may also include an odor absorbent material, for example activated charcoal, carbon fiber cloth or Vitec Carbotec-RT Q2003073 foam, or the like. For example, an odor absorbent material may form a layer of the filter element **130** or may be sandwiched between microporous hydrophobic membranes within the filter element.
(68) The filter element **130** thus enables gas to be exhausted through the orifice **145**. Liquid, particulates and pathogens however are contained in the dressing.
(69) In FIG. **1**B, an embodiment of the wound dressing **100** is illustrated which comprises spacer elements **152**, **153** in conjunction with the port **150** and the filter **130**. With the addition of such spacer elements **152**, **153**, the port **150** and filter **130** may be supported out of direct contact with the absorbent layer **110** and/or the transmission layer **105**. The absorbent layer **110** may also act as an additional spacer element to keep the filter **130** from contacting the transmission layer **105**. Accordingly, with such a configuration contact of the filter **130** with the transmission layer **105** and wound fluids during use may thus be minimized. As contrasted with the embodiment illustrated in FIG. **1**A, the aperture **146** through the absorbent layer **110** may not necessarily need to be as large or larger than the port **150**, and would thus only need to be large enough such that an air path can be maintained from the port to the transmission layer **105** when the absorbent layer **110** is saturated with wound fluids.
(70) In particular for embodiments with a single port **150** and through hole, it may be preferable for the port **150** and through hole to be located in an off-center position as illustrated in FIGS. **1**A-B and in FIG. **2**. Such a location may permit the dressing **100** to be positioned onto a patient such that the port **150** is raised in relation to the remainder of the dressing **100**. So positioned, the port **150** and the filter **130** may be less likely to come into contact with wound fluids that could prematurely occlude the filter **130** so as to impair the transmission of negative pressure to the wound site.
(71) FIG. **11** shows a plan view of a suction port **150** according to some embodiments of the invention. The suction port comprises a sealing surface **152** for sealing the port to a wound dressing, a connector portion **154** for connecting the suction port **150** to a source of negative pressure, and a hemispherical body portion **156** disposed between the sealing surface **152** and the connector portion **154**. Sealing surface **152** comprises a flange that provides a substantially flat area to provide a good seal when the port **150** is sealed to the cover layer **140**. Connector portion **154** is arranged to be coupled to the external source of negative pressure via a

length of tube **220**.

(72) According to some embodiments, the filter element **130** forms part of the bacterial barrier over the wound site, and therefore it is important that a good seal is formed and maintained around the filter element. However, it has been determined that a seal formed by adhering the filter element **130** to the cover layer **140** is not sufficiently reliable. This is a particular problem when a moisture vapor permeable cover layer is used, as the water vapor transpiring from the cover layer **140** can affect the adhesive, leading to breach of the seal between the filter element and the cover layer. Thus, according to some embodiments of the invention an alternative arrangement for sealing the filter element **130** to stop liquid from entering the connector portion **154** is employed.

(73) FIG. **12** illustrates a cross section through the suction port **150** of FIG. **11** according to some embodiments of the invention, the line A-A in FIG. **11** indicating the location of the cross section. In the suction port of FIG. **12**, the suction port **150** further comprises filter element **130** arranged within the body portion **156** of the suction port **150**. A seal between the suction port **150** and the filter element **130** is achieved by molding the filter element within the body portion of the suction port **150**.

(74) FIG. **13** illustrates a cross section through the suction port **150** of FIG. **11** according to some embodiments of the invention. In the suction port of FIG. **13**, the filter element **130** is sealed to the sealing surface **152** of the suction port **150**. The filter element may be sealed to the sealing surface using an adhesive or by welding the filter element to the sealing surface.

(75) By providing the filter element **130** as part of the suction port **150**, as illustrated in FIGS. **12** and **13**, the problems associated with adhering the filter element to the cover layer **140** are avoided allowing a reliable seal to be provided. Furthermore, providing a sub-assembly having the filter element **130** included as part of the suction port **150** allows for simpler and more efficient manufacture of the wound dressing **100**.

(76) While the suction port **150** has been described in the context of the wound dressing **100** of FIG. **1**, it will be understood that the embodiments of FIGS. **12** and **13** are applicable to any wound dressing for applying a negative pressure to a wound, wherein wound exudate drawn from the wound is retained within the dressing. According to some embodiments of the invention, the suction port **150** may be manufactured from a transparent material in order to allow a visual check to be made by a user for the ingress of wound exudate into the suction port **150**.

(77) The wound dressing **100** and its methods of manufacture and use as described herein may also incorporate features, configurations and materials described in the following patents and patent applications incorporated by reference in their entireties: U.S. Pat. Nos. 7,524,315, 7,708,724, and 7,909,805; U.S. Patent Application Publication Nos. 2005/0261642, 2007/0167926, 2009/0012483, 2009/0254054, 2010/0160879, 2010/0160880, 2010/0174251, 2010/0274207, 2010/0298793, 2011/0009838, 2011/0028918, 2011/0054421, and 2011/0054423; as well as U.S. application Ser. No. 12/941,390, filed Nov. 8, 2010, Ser. No. 29/389,782, filed Apr. 15, 2011, and Ser. No. 29/389,783, filed Apr. 15, 2011. From these incorporated by reference patents and patent applications, features, configurations, materials and methods of manufacture or use for similar components to those described in the present disclosure may be substituted, added or implemented into embodiments of the present application.

(78) In operation the wound dressing **100** is sealed over a wound site forming a wound cavity. A pump unit (illustrated in FIG. **28** and described in further detail below) applies a negative pressure at a connection portion **154** of the port **150** which is communicated through the orifice **145** to the transmission layer **105**. Fluid is drawn towards the orifice through the wound dressing from a wound site below the wound contact layer **102**. The fluid moves towards the orifice through the transmission layer **105**. As the fluid is drawn through the transmission layer **105** wound exudate is absorbed into the absorbent layer **110**.

(79) Turning to FIG. **2** which illustrates a wound dressing **100** in accordance with an embodiment of the present invention one can see the upper surface of the cover layer **140** which extends outwardly away from a centre of the dressing into a border region **200** surrounding a central raised region **201** overlying the transmission layer **105** and the absorbent layer **110**. As indicated in FIG. **2** the general shape of the wound dressing is rectangular with rounded corner regions **202**. It will be appreciated that wound dressings according to other embodiments of the present invention can be shaped differently such as square, circular or elliptical dressings, or the like.

(80) The wound dressing **100** may be sized as necessary for the size and type of wound it will be used in. In some embodiments, the wound dressing **100** may measure between 20 and 40 cm on its long axis, and between 10 to 25 cm on its short axis. For example, dressings may be provided in sizes of 10×20 cm, 10×30 cm, 10×40 cm, 15×20 cm, and 15×30 cm. In some embodiments, the wound dressing **100** may be a square-

shaped dressing with sides measuring between 15 and 25 cm (e.g., 15×15 cm, 20×20 cm and 25×25 cm). The absorbent layer **110** may have a smaller area than the overall dressing, and in some embodiments may have a length and width that are both about 3 to 10 cm shorter, more preferably about 5 cm shorter, than that of the overall dressing **100**. In some rectangular-shape embodiments, the absorbent layer **110** may measure between 10 and 35 cm on its long axis, and between 5 and 10 cm on its short axis. For example, absorbent layers may be provided in sizes of 5.6×15 cm or 5×10 cm (for 10×20 cm dressings), 5.6×25 cm or 5×20 cm (for 10×30 cm dressings), 5.6×35 cm or 5×30 cm (for 10×40 cm dressings), 10×15 cm (for 15×20 cm dressings), and 10×25 cm (for 15×30 cm dressings). In some square-shape embodiments, the absorbent layer **110** may have sides that are between 10 and 20 cm in length (e.g., 10×10 cm for a 15×15 cm dressing, 15×15 cm for a 20×20 cm dressing, or 20×20 cm for a 25×25 cm dressing). The transmission layer **105** is preferably smaller than the absorbent layer, and in some embodiments may have a length and width that are both about 0.5 to 2 cm shorter, more preferably about 1 cm shorter, than that of the absorbent layer. In some rectangular-shape embodiments, the transmission layer may measure between 9 and 34 cm on its long axis and between 3 and 5 cm on its short axis. For example, transmission layers may be provided in sizes of 4.6×14 cm or 4×9 cm (for 10×20 cm dressings), 4.6×24 cm or 4×19 cm (for 10×30 cm dressings), 4.6×34 cm or 4×29 cm (for 10×40 cm dressings), 9×14 cm (for 15×20 cm dressings), and 9×24 cm (for 15×30 cm dressings). In some square-shape embodiments, the transmission layer may have sides that are between 9 and 19 cm in length (e.g., 9×9 cm for a 15×15 cm dressing, 14×14 cm for a 20×20 cm dressing, or 19×19 cm for a 25×25 cm dressing).

(81) It will be understood that according to embodiments of the present invention the wound contact layer is optional. This layer is, if used, porous to water and faces an underlying wound site. A transmission layer **105** such as an open celled foam, or a knitted or woven spacer fabric is used to distribute gas and fluid removal such that all areas of a wound are subjected to equal pressure. The cover layer together with the filter layer forms a substantially liquid tight seal over the wound. Thus when a negative pressure is applied to the port **150** the negative pressure is communicated to the wound cavity below the cover layer. This negative pressure is thus experienced at the target wound site. Fluid including air and wound exudate is drawn through the wound contact layer and transmission layer **105**. The wound exudate drawn through the lower layers of the wound dressing is dissipated and absorbed into the absorbent layer **110** where it is collected and stored. Air and moisture vapor is drawn upwards through the wound dressing through the filter layer and out of the dressing through the suction port. A portion of the water content of the wound exudate is drawn through the absorbent layer and into the cover layer **140** and then evaporates from the surface of the dressing.

(82) As discussed above, when a negative pressure is applied to a wound dressing sealed over a wound site, fluids including wound exudate are drawn from the wound site and through the transmission layer **105** towards the orifice **145**. Wound exudate is then drawn into the absorbent layer **110** where it is absorbed. However, some wound exudate may not be absorbed and may move to the orifice **145**. Filter element **130** provides a barrier that stops any liquid in the wound exudate from entering the connection portion **154** of the suction port **150**. Therefore, unabsorbed wound exudate may collect underneath the filter element **130**. If sufficient wound exudate collects at the filter element, a layer of liquid will form across the surface of filter element **130** and the filter element will become blocked as the liquid cannot pass through the filter element **130** and gases will be stopped from reaching the filter element by the liquid layer. Once the filter element becomes blocked, negative pressure can no longer be communicated to the wound site, and the wound dressing must be changed for a fresh dressing, even though the total capacity of the absorbent layer has not been reached.

(83) In a preferred embodiment, the port **150**, along with any aperture **146** in the absorbing layer **110** situated below it, generally aligns with the mid-longitudinal axis A-A illustrated in FIG. **2**. Preferably, the port **150** and any such aperture **146** are situated closer to one end of the dressing, contrasted with a central position. In some embodiments, the port may be located at a corner of the dressing **100**. For example, in some rectangular embodiments, the port **150** may be located between 4 and 6 cm from the edge of the dressing, with the aperture **146** located 2 to 3 cm from the edge of the absorbent layer. In some square embodiments, the port **150** may be located between 5 to 8 cm from the corner of the dressing, with the aperture **146** located 3 to 5 cm from the corner of the absorbent layer.

(84) Certain orientations of the wound dressing may increase the likelihood of the filter element **130** becoming blocked in this way, as the movement of the wound exudate through the transmission layer may be aided by the effect of gravity. Thus, if due to the orientation of the wound site and wound dressing, gravity acts to increase the rate at which wound exudate is drawn towards the orifice **145**, the filter may become blocked with wound exudate more quickly. Thus, the wound dressing would have to be changed more

frequently and before the absorbent capacity of the absorbent layer **110** has been reached.

(85) In order to avoid the premature blocking of the wound dressing **100** by wound exudate drawn towards the orifice **145** some embodiments of the invention include at least one element configured to reduce the rate at which wound exudate moves towards the orifice **145**. The at least one element may increase the amount of exudate that is absorbed into the absorbent layer before reaching the orifice **145** and/or may force the wound exudate to follow a longer path through the dressing before reaching the orifice **145**, thereby increasing the time before the wound dressing becomes blocked.

(86) FIG. **3** shows a plan view of a wound dressing including baffle elements that reduce the rate at which wound exudate moves towards the orifice according to one embodiment of the invention. The wound dressing illustrated in FIG. **3** is similar to that shown in FIGS. **1** and **2**, but includes a number of baffle elements **310** disposed across the central raised region **201**. The baffle elements **310** form barriers in the central region of the dressing, which arrest the movement of wound exudate towards the orifice.

(87) Embodiments of baffle elements that may be used in the wound dressing described herein are preferably at least partly flexible, so as to permit the wound dressing to flex and conform with the skin of the patient surrounding the wound site. When so present in the wound dressing, the baffle elements are preferably constructed so as to at least partially prevent liquid from flowing directly to the wound dressing port or orifice and its associated filter, if so provided. The baffle elements thus increase the distance that liquids may require to reach the port, which may help in absorbing these fluids into the absorbent or superabsorbent material of the wound dressing.

(88) According to some embodiments of the invention, the baffle element may comprise a sealing region in which the absorbent layer **110** and transmission layer **105** are absent and cover layer **140** is sealed to the wound contact layer **101**. Thus, the baffle element presents a barrier to the motion of the wound exudate, which must therefore follow a path that avoids the baffle element. Thus the time taken for the wound exudate to reach the orifice is increased.

(89) In some embodiments, the baffle elements may be an insert of a substantially non-porous material, for example a closed-cell polyethylene foam, placed inside the dressing. In some cases, it may be preferable to place such an inserted baffle element in a sealing region where one or more of the absorbent layer **110** and/or transmission layer **105** are absent. A sealant, for example a viscous curing sealant such as a silicone sealant, could be placed or injected as a thin strip so as to form a baffle element that is substantially liquid impermeable. Such a baffle element could be placed or injected into a region of the transmission layer **105** and/or absorbent layer **110**, or also a sealing region where the absorbent layer **110** and/or transmission layer **105** are absent.

(90) FIG. **6** illustrates a wound dressing including a baffle element according to a further embodiment of the invention. A single baffle element **610** provides a cup shaped barrier between the bulk of the absorbent layer **110** and the orifice **145**. Thus wound exudate that is initially drawn from the wound site within the region defined by the baffle element **610**, must follow a path around the outside of the cup shaped barrier to reach the orifice **145**. As will be recognized, the baffle element **610** reduces the effect of gravity on reducing the time taken for the wound exudate to move to the orifice **145**, as for most orientations of the wound dressing at least a part of the path taken by the wound exudate will be against the force of gravity.

(91) The embodiments of FIGS. **3** and **6** have been described with respect to a wound dressing having a structure as shown in FIG. **1**. However, it will be understood that the baffle elements could equally be applied to a wound dressing in which the transmission layer **105** was absent.

(92) FIG. **4** shows a plan view of a wound dressing including the at least one element according to one embodiment of the invention in which a number of baffle elements **410** are provided that extend across the width of the central region **201** of the wound dressing, with further baffle elements **412** formed in a semi-circular path around the orifice **145**.

(93) FIG. **5** illustrates the configuration of baffle elements **410** according to some embodiments of the invention. The baffle element comprises a channel of absorbent material **510** underlying the transmission layer **105**. A channel in the absorbent layer **110** is located over the baffle element **410** so that the transmission layer is in contact with the cover layer **140** in the region of the baffle element **410**. Thus, wound exudate that is moving along a lower surface of the transmission layer **105**, and has therefore not been drawn into absorbent layer **110**, will come into contact with and be absorbed by the channel of absorbent material **510**.

(94) Alternatively, or additionally, baffle elements may comprise one or more channels provided in the surface of the transmission layer **105** underlying and abutting the absorbent layer **110**. In use, when negative pressure is applied to the wound dressing, the absorbent layer **110** will be drawn into the channel. The

channel in the transmission layer may have a depth substantially equal to the depth of the transmission layer, or may have a depth less than the depth of the transmission layer. The dimensions of the channel may be chosen to ensure that the channel is filled by the absorbent layer **110** when negative pressure is applied to the wound dressing. According to some embodiments, the channel in the transmission layer comprises a channel of absorbent material in the transmission layer **105**.

(95) The baffle elements may be formed into a range of shapes and patterns, for example FIGS. **14**A to **14**L illustrate wound dressings having a number of different exemplifying configurations of baffle elements. FIG. **14**A illustrates a linear baffle element in a vertical configuration aligned in the direction of the port or orifice. FIG. **14**B illustrates an X-shaped baffle element. FIGS. **14**C-E illustrate embodiments of wound dressings with multiple baffle elements, aligned in a generally diagonal, horizontal, or vertical manner.

(96) FIG. **14**F illustrates baffle elements arranged in a six-armed starburst configuration, with a center portion left open. FIG. **14**G illustrates a W-shaped baffle element on the wound dressing in a position distal to the port or orifice. In FIG. **14**H, an 3-by-3 array of X-shaped baffle elements is provided on the wound dressing, although it will be understood that more or less X-shaped baffle elements may be used. FIG. **14**I shows an embodiment with a plurality of rectangular baffle elements, and wherein one or more baffle elements are located underneath the port in the wound dressing. FIGS. **14**J-K illustrate wound dressing embodiments with longer diagonal and horizontal baffle elements. In FIG. **14**L, rectangular baffle elements are present on this embodiment of a wound dressing, wherein the baffle elements are of different sizes.

(97) According to some embodiments of the invention, the at least one element comprises an array of vias, or troughs, in the transmission layer **105**. FIG. **15** illustrates a transmission layer **105** that is perforated with diamond shaped vias **210**. The vias **210** are arranged such that no linear pathway exists through the pattern of vias that does not intersect with one or more of the vias **210**.

(98) When negative pressure is applied to the wound dressing, the absorbent layer **110** is drawn into the vias **210**, increasing the area of the absorbent layer that comes into contact with wound exudate being drawn through the transmission layer **105**. Alternatively, the vias **210** may be filled with further absorbent material for absorbing wound exudate being drawn through the transmission layer **105**. The vias may extend through the depth of the transmission layer **105**, or may extend through only part of the transmission layer.

(99) Wound exudate moving through the transmission layer **105** under the influence of gravity will fall through the transmission layer in a substantially linear manner. Any such linear pathways will, at some point, intersect with one of the vias **210**, and thus the exudate will be brought into contact with absorbent material within the vias **210**. Wound exudate coming into contact with absorbent material will be absorbed, stopping the flow of the wound exudate through the transmission layer **105**, and reducing the amount of unabsorbed wound exudate that may otherwise pool around the orifice. It will be appreciated that the vias are not limited to diamond shapes, and that any pattern of vias may be used. Preferably, the vias will be arranged to ensure that all linear paths through the transmission layer **105** intersect with at least one via. The pattern of vias may be chosen to minimize the distance that wound exudate is able to travel though the transmission layer before encountering a via and being absorbed.

(100) FIG. **7** illustrates a wound dressing in accordance with some embodiments of the invention in which the at least one element comprises an air channel **710** connecting the central region **201** of the wound dressing to the orifice **145**. In the embodiment of FIG. **7**, the air channel **710** extends from an edge region of the transmission layer **105** and connects the transmission layer to the orifice **145**.

(101) In use, wound exudate is drawn towards the orifice **145** by the application of negative pressure at the suction port **150**. However, the air channel **710** present a relatively long serpentine path to be followed by the wound exudate before it reaches the orifice **145**. This long path increases the time that negative pressure can be applied to the dressing before wound exudate traverses the distance between the transmission layer and the orifice and blocks the filter element **130**, thereby increasing the time the dressing can be in use before it must be replaced.

(102) FIG. **8** illustrates a wound dressing in accordance with one embodiment of the invention in which the at least one element comprises air channels **810** and **812** connecting the central region **201** of the wound dressing to the orifice **145**. Channels **810** and **812** are coupled to the transmission layer at substantially opposite corners of the central region **201**.

(103) The wound dressing shown in FIG. **8** reduces the effect of gravity on the time taken for the orifice to become blocked. If the wound dressing is in an orientation in which wound exudate moves under the influence of gravity towards the edge region of the transmission layer connected to air channel **810**, the effect of gravity will be to move wound exudate away from the edge region of the transmission layer coupled to air

channel **812**, and vice versa. Thus, the embodiment of FIG. **8** provides alternative air channels for coupling the negative pressure to the transmission layer such that, should one air channel become blocked a remaining air channel should remain open and able to communicate the negative pressure to the transmission layer **105**, thereby increasing the time before negative pressure can no longer be applied to the wound dressing and the dressing must be changed.

(104) Further embodiments of the invention may comprise greater numbers of air channels connecting the transmission layer **105** to the orifice.

(105) According to some embodiments of the invention, two or more orifices may be provided in the cover layer **140** for applying the negative pressure to the wound dressing. The two or more orifices can be distributed across the cover layer **140** such that if one orifice becomes blocked by wound exudate due to the wound dressing being in a particular orientation, at least one remaining orifice would be expected to remain unblocked. Each orifice is in fluid communication with a wound chamber defined by the wound dressing, and is therefore able to communicate the negative pressure to the wound site.

(106) FIG. **9** illustrates a wound dressing in accordance with a further embodiment of the invention. The wound dressing of FIG. **9** is similar to that of FIGS. **1**A-B but includes two orifices **145** and **845** provided in the cover layer **140**. A fluid communication passage connects the two orifices such that a negative pressure applied to one of the orifices is communicated to the remaining orifice via the fluid communication passage. The orifices **145**, **845** are located in opposite corner regions of the cover layer **140**. The fluid communication passage is formed using a flexible molding **910** on the upper surface of the cover layer **140**. It will be appreciated that the flexible molding may be formed from other suitable means for example a strip of transmission or open porous foam layer placed on the cover layer **140** between the orifices **145** and **845** and a further film welded or adhered over the strip thus sealing it to the cover layer and forming a passageway through the foam. A conduit may then be attached in a known manner to the sealing film for application of negative pressure.

(107) In use, the wound dressing having two orifices is sealed over a wound site to form a wound cavity and an external source of negative pressure is applied to one of the orifices **145**, **845**, and the negative pressure will be communicated to the remaining orifice via the fluid communication passage. Thus, the negative pressure is communicated via the two orifices **145**, **845** to the transmission layer **105**, and thereby to the wound site. If one of the orifices **145**, **845** becomes blocked due to wound exudate collecting at the orifice under the influence of gravity, the remaining orifice should remain clear, allowing negative pressure to continue to be communicated to the wound site. According to some embodiments, the transmission layer **105** may be omitted, and the two orifices will communicate the negative pressure to the wound site via the absorbent layer **110**.

(108) FIG. **10** illustrates a side view of the fluid communication passage of the embodiment of FIG. **9**. Molding **910** is sealed to the top surface of the cover layer **140**, and covering orifices **145** and **845**. Gas permeable liquid impermeable filter elements **130** are provided at each orifice. The molding **910** is coupled to an external source of negative pressure via a tube element **220**.

(109) According to some embodiments, a single filter element may be used extending underneath the length of the fluid communication passage and the two orifices. While the above example embodiment has been described as having two orifices, it will be understood that more than two orifices could be used, the fluid communication passage allowing the negative pressure to be communicated between the orifices.

(110) FIG. **16** illustrates an alternative arrangement in which a single elongate orifice **350** is provided in the cover layer **140**. First and second ends **355**, **356** of the orifice **350** are located in opposite corner regions of the cover layer **140**. A flexible molding **360** is sealed around the orifice **350** and allows negative pressure to be communicated through the cover layer **140** along the length of the orifice **350**. The flexible molding **360** may be formed by any suitable means as described above in relation to flexible molding **910**.

(111) In use, the wound dressing is sealed over a wound site to form a wound cavity and an external source of negative pressure is applied to the orifice. If, due to the orientation of the wound dressing, wound exudate moves under the influence of gravity to collect around one end **355** of the orifice **350**, a portion of the orifice **350** near to the end **355** will become blocked. However, a portion of the orifice near to the remaining end **356** should remain clear, allowing continued application of negative pressure to the wound site.

(112) As still further options the dressing can contain anti-microbial e.g. nanocrystalline silver agents on the wound contact layer and/or silver sulphur diazine in the absorbent layer. These may be used separately or together. These respectively kill micro-organisms in the wound and micro-organisms in the absorption matrix. As a still further option other active components, for example, pain suppressants, such as ibuprofen,

may be included. Also agents which enhance cell activity, such as growth factors or that inhibit enzymes, such as matrix metalloproteinase inhibitors, such as tissue inhibitors of metalloproteinase (TIMPS) or zinc chelators could be utilized. As a still further option odor trapping elements such as activated carbon, cyclodextrine, zeolite or the like may be included in the absorbent layer or as a still further layer above the filter layer.

(113) FIG. **17** illustrates a first, upper surface **1700** and a further, lower surface **1702** of a transmission layer **105** according to an embodiment of the present invention. In the embodiment illustrated in FIG. **17** fibers **1703** of a woven layer extend between the first surface **1700** and the further surface **1702**. It will be appreciated that according to further embodiments of the present invention if a foam layer is used as a transmission layer **105** the connected strands forming the foam will act as spacer elements. As illustrated in FIG. **17** in a relaxed mode of operation, that is to say when in use, no negative pressure is applied to the wound dressing or negative pressure is applied to the wound dressing but no external force acts on the wound dressing then the fibers **1703** extend substantially perpendicular to the upper and lower surfaces keeping the surfaces in a spaced apart substantially parallel configuration.

(114) FIG. **18** illustrates the transmission layer **105** when an external force is exerted on the outside of the dressing. The external force can be a compressive force indicated by arrow A and/or a lateral force illustrated by arrow B in FIG. **18**. As indicated either a compressive force or a lateral force acts to cause the fibers **1703** to lean to one side. This causes the upper and lower surfaces to become laterally offset with respect to each other as well as causing the thickness of the layer to reduce from a separation distance r indicated in FIG. **17** in a relaxed mode of operation to a compression distance c illustrated in FIG. **18**. The reduction in thickness effectively provides some “give” in the dressing even when the dressing is subject to negative pressure. It will be appreciated that the forces acting on the dressing may occur throughout the whole of the surface area of the dressing or only in one or more particular regions. In such a situation regions of the dressing can be in a relaxed mode of operation and further regions can be in a compressed mode of operation. As illustrated in FIG. **18** when a force is exerted on the transmission layer the fibers separating the upper and lower surfaces tend to lean to one side sharing a common lean angle.

(115) Throughout this specification reference will be made to a relaxed mode of operation and a forced mode of operation. It is to be understood that the relaxed mode of operation corresponds to a natural state of the material either when no negative pressure is applied or when negative pressure is applied. In either situation no external force, caused for example by motion of a patient or an impact is in evidence. By contrast a forced mode of operation occurs when an external force whether compressive, lateral or other is brought to bear upon the wound dressing. Such forces can cause serious damage/prevent healing or a wound.

(116) FIG. **19** illustrates how certain embodiments of the present invention can also operate to offset load forces. As illustrated in FIG. **19** if a force is exerted over a contact area **1900** in an upper surface **1700** of the transmission layer **105** then this force is transmitted across and through the transmission layer and is exerted over a larger dissipation area **1901** against an underlying wound site. In the case of use of a 3D knit as a transmission layer this is because the relatively stiff spacer elements provide at least some lateral stiffness to the layer.

(117) FIG. **20** illustrates the transmission layer **105** and absorbent layer **110** of some embodiments in more detail. The absorbent layer **110** is located proximate to the upper surface **1700** of the transmission layer **105** and is unbonded thereto according to certain embodiments of the present invention. When unbonded the absorbent layer **110** is also able to move laterally with respect to the underlying transmission layer when a lateral or shear force is applied to the wound dressing. Also the absorbent layer is able to further compress when a compressive force illustrated in FIG. **21** acts on the wound dressing. As illustrated in FIG. **21** the absorbent layer **110** decreases in thickness under a compressive force from a non-compressed thickness x illustrated in FIG. **20** to a compressed distance y illustrated in FIG. **21**. The compressive force also acts to offset the upper and lower surfaces of the transmission layer as described above thus enhancing the “give” of the dressing. The ability for an upper surface **2201** to translate laterally with respect to a lower surface **2202** of the absorbent layer under a lateral or shearing force exerted on the wound dressing is illustrated in more detail in FIG. **22**. This lateral motion causes the thickness x of the absorbent layer **110** to reduce and the upper surface and lower surface of the absorbent layer to be offset with respect to each other. This effect can itself be sufficient to prevent shear forces exerted on the whole or part of the wound dressing from being transferred to an underlying wound bed. As can the corresponding effect in the transmission layer. However a combination enhances the cushioning effect. If the wound bed comprises a skin graft region the reduction of shear forces can be particularly advantageous.

(118) It is to be noted that in use the dressing may be used "up-side down", at an angle or vertical. References to upper and lower are thus used for explanation purposes only.
(119) FIG. **23** illustrates a cross-section of a portion of an embodiment of a dressing shown in FIGS. **1**A-**2**. In particular, FIG. **23** illustrates a magnified view of the wound contact layer **102** which includes a lower surface **101** and multiple perforations **104** formed as through holes. An upper surface **104** of the wound contact layer abuts a first layer **300** of the transmission layer **105**. A further, upper, layer **301** of the transmission layer **105** is spaced apart from the first layer. The first and further layers of the transmission layer are kept apart in a spaced apart relationship by multiple mono-filament fiber spacers **302** which act as resilient flexible pillars separating the two layers of the transmission layer. The upper layer **301** of the transmission layer is adjacent a lower surface of the absorbent **110** which, for example, is formed as a pad of fibrous cellulose material interspaced with super-absorbent particulate matter.
(120) FIG. **24** illustrates the lower layer of the 3D fabric transmission layer in more detail. The 3D fabric layer **105** is formed as a lower and upper knitted layer given a loft by the knitted structure. Rows of the knitted stitches may be referred to as a course of stitches. Columns of stitches may be referred to as a whale. A single monofilament fiber is knitted into the 3D fabric to form the multiple separating strands.
(121) As illustrated in FIG. **24** there are apertures or openings formed between interlocked stitches in the lower layer of the transmission layer **105**. In use, wound exudate including liquid and semi-solid e.g. viscous slurry, suspensions of biological debris or the like and solid material will pass upwards through the perforations **104** in the wound contact layer and through the openings in the inter knitted structure of the first layer **300** of the transmission layer. The openings between the interconnected stitches have an average open area ranging from around 250 microns to 450 microns. The particular open area in the first layer of the transmission layer will be determined by the materials and method of manufacture of the lower layer. FIG. **25** illustrates how an open area of openings in the further layer above the first layer (that is to say further away from the wound) can include openings which have a greater open area than the openings in the lower layer. In this way as wound exudate which includes semi-solid and solid matter moves from the wound bed at the wound site upwards into the wound dressing any particulate matter which is of a size small enough to pass through the relative small openings **400** in the lower layer will certainly be able to pass through the larger area openings **501** in the upper area. This helps avoid debris in the form of solid material collecting in the interstitial region between the monofilament fibers between the upper and lower layer. As shown in FIG. **25**, the upper layer **301** may include openings **500** similar to the openings **400** in the lower layer **300**. However, during the knitting process the upper surface is knitted so that larger open area openings **501** are interspersed across the whole surface of the upper layer. As illustrated in FIG. **25** the larger open area openings **501** can have an open range considerably larger (shown between 700 to 800 microns). The lower layer **300** thus acts to some extent as a filtering layer having openings **400** which enable gas and liquid to pass freely therethrough but to prevent solid and semi-solid particulate matter which is too large from passing in to the interstitial region in the transmission layer **105**. This helps keep a flowpath along the transmission layer open.
(122) By providing openings in an upper layer in the transmission layer which have a greater open area than any openings in the lower area build-up of solid particulate matter in the interstitial region between the upper and lower layers of the transmission layer is avoided since any solid or semi-solid matter will flow along the channel and eventually be enabled to pass upwards through the larger openings where the material is taken up by the super-absorber/absorbent material.
(123) The absorbent layer **110** holds liquid collected during the application of negative pressure therapy. By having this layer in fluid communication with, and preferably in contact with, the layer of the transmission layer, the region of the transmission layer **105** is kept at a moist environment. This helps avoid build-up and crusting of the exudate during use.
(124) FIG. **26** illustrates an alternative material which could be utilized as the transmission layer in a wound dressing. In particular, FIG. **26** illustrates a lower surface of a 3D knit material which may be utilized as the transmission layer. Openings **600** are formed in the surface which enables wound exudate and air to pass from the wound through a wound contact layer which would be located on the surface shown in FIG. **6** and through those openings. FIG. **27** illustrates an upper surface of the material shown in FIG. **26** and illustrates how larger openings **700** may be formed in the upper surface.
(125) Whilst certain embodiments of the present invention have so far been described in which the transmission layer is formed as a 3D knit layer, e.g., two layers spaced apart by a monofilament layer, it will be appreciated that certain embodiments of the present invention are not restricted to the use of such a material. In some embodiments, as an alternative to such a 3D knit material one or more layers of a wide

variety of materials could be utilized. In each case, according to embodiments of the present invention, the openings presented by layers of the transmission layer are wider and wider as one moves away from the side of the dressing which, in use will be located proximate to the wound. In some embodiments, the transmission layer may be provided by multiple layers of open celled foam. In some embodiments, the foam is reticulated open cell foam. Preferably, the foam is hydrophilic or able to wick aqueous based fluids. The pore size in each layer is selected so that in the foam layer most proximate to the wound side in use the pores have a smallest size. If only one further foam layer is utilized that includes pore sizes which are greater than the pore sizes of the first layer. This helps avoid solid particulate being trapped in the lower layer which thus helps maintain the lower layer in an open configuration in which it is thus able to transmit air throughout the dressing. In certain embodiments, two, three, four or more foam layers may be included. The foam layers may be integrally formed, for example, by selecting a foam having a large pore size and then repeatedly dipping this to a lesser and lesser extent into material which will clog the pores or alternatively, the transmission layer formed by the multiple foam layers may be provided by laminating different types of foam in a layered arrangement or by securing such layers of foam in place in a known manner.

(126) According to certain embodiments of the present invention, the transmission layer is formed by multiple layers of mesh instead of foam or 3D knit materials. For example, fine gauze mesh may be utilized for a wound facing side of the transmission layer and a Hessian mesh having a larger pore size may be located on a distal side of the gauze mesh facing away from the wound in use. The one, two, three or more layers of mesh can be secured together in an appropriate manner, such as being stitched or adhered together or the like. The resultant mat of fibers provides a transmittal layer through which air can be transmitted in the dressing but by selecting the opening sizes in the meshes as one moves through the dressing away from the wound contact side, the accumulation of solid particulate matter in lower layers can be avoided.

(127) FIG. **28** illustrates an embodiment of a TNP wound treatment comprising a wound dressing **100** in combination with a pump **800**. Here, the dressing **100** may be placed over a wound as described previously, and a conduit **220** may then be connected to the port **150**, although in some embodiments the dressing **100** may be provided with at least a portion of the conduit **220** preattached to the port **150**. Preferably, the dressing **100** is provided as a single article with all wound dressing elements (including the port **150**) pre-attached and integrated into a single unit. The wound dressing **100** may then be connected, via the conduit **220**, to a source of negative pressure such as the pump **800**. Preferably, the pump **800** is miniaturized and portable, although larger conventional pumps may also be used with the dressing **100**. In some embodiments, the pump **800** may be attached or mounted onto or adjacent the dressing **100**. A connector **221** may also be provided so as to permit the conduit **220** leading to the wound dressing **100** to be disconnected from the pump, which may be useful for example during dressing changes.

(128) FIGS. **29**A-D illustrate the use of an embodiment of a TNP wound treatment system being used to treat a wound site on a patient. FIG. **29**A shows a wound site **190** being cleaned and prepared for treatment. Here, the healthy skin surrounding the wound site **190** is preferably cleaned and excess hair removed or shaved. The wound site **190** may also be irrigated with sterile saline solution if necessary. Optionally, a skin protectant may be applied to the skin surrounding the wound site **190**. If necessary, a wound packing material, such as foam or gauze, may be placed in the wound site **190**. This may be preferable if the wound site **190** is a deeper wound.

(129) After the skin surrounding the wound site **190** is dry, and with reference now to FIG. **29**B, the wound dressing **100** may be positioned and placed over the wound site **190**. Preferably, the wound dressing **100** is placed with the wound contact layer **102** over and/or in contact with the wound site **190**. In some embodiments, an adhesive layer is provided on the lower surface **101** of the wound contact layer **102**, which may in some cases be protected by an optional release layer to be removed prior to placement of the wound dressing **100** over the wound site **190**. Preferably, the dressing **100** is positioned such that the port **150** is in a raised position with respect to the remainder of the dressing **100** so as to avoid fluid pooling around the port. In some embodiments, the dressing **100** is positioned so that the port **150** is not directly overlying the wound, and is level with or at a higher point than the wound. To help ensure adequate sealing for TNP, the edges of the dressing **100** are preferably smoothed over to avoid creases or folds.

(130) With reference now to FIG. **29**C, the dressing **100** is connected to the pump **800**. The pump **800** is configured to apply negative pressure to the wound site via the dressing **100**, and typically through a conduit. In some embodiments, and as described above in FIG. **28**, a connector may be used to join the conduit from the dressing **100** to the pump **800**. Upon the application of negative pressure with the pump **800**, the dressing **100** may in some embodiments partially collapse and present a wrinkled appearance as a result of the

evacuation of some or all of the air underneath the dressing **100**. In some embodiments, the pump **800** may be configured to detect if any leaks are present in the dressing **100**, such as at the interface between the dressing **100** and the skin surrounding the wound site **190**. Should a leak be found, such leak is preferably remedied prior to continuing treatment.

(131) Turning to FIG. **29**D, additional fixation strips **195** may also be attached around the edges of the dressing **100**. Such fixation strips **195** may be advantageous in some situations so as to provide additional sealing against the skin of the patient surrounding the wound site **190**. For example, the fixation strips **195** may provide additional sealing for when a patient is more mobile. In some cases, the fixation strips **195** may be used prior to activation of the pump **800**, particularly if the dressing **100** is placed over a difficult to reach or contoured area.

(132) Treatment of the wound site **190** preferably continues until the wound has reached a desired level of healing. In some embodiments, it may be desirable to replace the dressing **100** after a certain time period has elapsed, or if the dressing is full of wound fluids. During such changes, the pump **800** may be kept, with just the dressing **100** being changed.

(133) Throughout the description and claims of this specification, the words "comprise" and "contain" and variations of the words, for example "comprising" and "comprises", mean "including but not limited to", and is they are not intended to (and does not) exclude other moieties, additives, components, integers or steps.

(134) Throughout the description and claims of this specification, the singular encompasses the plural unless the context otherwise requires. In particular, where the indefinite article is used, the specification is to be understood as contemplating plurality as well as singularity, unless the context requires otherwise.

(135) Features, integers, characteristics, compounds, chemical moieties or groups described in conjunction with a particular aspect, embodiment or example of the invention are to be understood to be applicable to any other aspect, embodiment or example described herein unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The invention is not restricted to the details of any foregoing embodiments. The invention extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

Claims

1. A transmission layer for use in a wound dressing, the transmission layer comprising: a knitted or woven spacer fabric comprising a three dimensional structure, the knitted or woven spacer fabric comprising a hydrophilic agent configured such that liquid enters the knitted or woven spacer fabric as soon as liquid contacts the knitted or woven spacer fabric, wherein the three dimensional structure comprises a top spacer fabric layer and a bottom spacer fabric layer separated from the top spacer fabric layer by a plurality of perpendicular spacer elements, and wherein the filament count in the top spacer fabric layer is greater than the filament count in the bottom spacer fabric layer.
2. The transmission layer of claim 1, further comprising a knitted layer of at least partially elastomeric material, the elastomeric material comprising polyester.
3. The transmission layer of claim 1, wherein the transmission layer is configured to allow transmission of liquid and gas away from a wound site into upper layers of a wound dressing.
4. The transmission layer of claim 1, wherein the top spacer fabric layer is spaced apart from the bottom spacer fabric layer by a relax distance in a relaxed mode of operation; and the plurality of spacer elements extend between the top spacer fabric layer and the bottom spacer fabric layer and, in a forced mode of operation, locatable whereby the top spacer fabric layer and the bottom spacer fabric layer are spaced apart by a compression distance less than the relax distance.
5. A wound dressing for use in topical negative pressure therapy, comprising the transmission layer of claim 1.
6. The wound dressing of claim 5, further comprising a layer of absorbent material above the transmission layer.
7. The wound dressing of claim 6, wherein the layer of absorbent material is not bonded to the transmission layer.
8. The wound dressing of claim 6, wherein the layer of absorbent material comprises super-absorbent

material.

9. The wound dressing of claim 5, further comprising: a moisture vapour permeable drape layer over the transmission layer; and a suction port configured to be connected to a source of negative pressure.

10. A process for preparing a transmission layer for use in a wound dressing, the process comprising: providing a transmission layer comprising a knitted or woven spacer fabric having a three dimensional structure comprising a top fabric layer and a bottom fabric layer separated from the top fabric layer by a plurality of perpendicular spacer elements, wherein the filament count in the top fabric layer is greater than the filament count in the bottom fabric layer, and washing the transmission layer in a hydrophilic agent such that liquid enters the knitted or woven spacer fabric as soon as liquid contacts the knitted or woven spacer fabric.

11. The process of claim 10, wherein the transmission layer comprises a knitted layer of at least partially elastomeric material comprising polyester.

12. The process of claim 10, wherein the process further comprises treating the transmission layer with a dry cleaning agent to remove manufacturing products such as mineral oils, fats and/or waxes that would otherwise interfere with the hydrophilic capabilities of the transmission layer.

13. The process of claim 10, wherein the top fabric layer is spaced apart from the bottom fabric layer by a relax distance in a relaxed mode of operation; and the plurality of spacer elements extend between the top fabric layer and the bottom fabric layer and, in a forced mode of operation, locatable whereby the top fabric layer and the bottom fabric layer are spaced apart by a compression distance less than the relax distance.

14. A transmission layer as treated by the process of claim 10.

15. A wound dressing for use in topical negative pressure therapy comprising a transmission layer treated by the process of claim 10.

16. A wound dressing comprising a transmission layer treated by the process of claim 10, further comprising a layer of absorbent material above the transmission layer.

17. The wound dressing of claim 16, wherein the layer of absorbent material is disposed proximate to transmission layer and unbonded thereto.

18. The wound dressing of claim 16, wherein the layer of absorbent material comprises super-absorbent material.

19. The wound dressing of claim 16, further comprising a moisture vapor permeable drape layer over the transmission layer, and a suction port connectable to a source of negative pressure.